

# Rank Before You Merge: A Stage Law for Training-Free Visual Token Compression in Merger-Equipped Vision-Language Models

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## Abstract

Native spatial mergers reduce visual-token count before the language model, but their effect on later token selection has not been isolated. We compare the same query-blind L2 selector immediately before and after the native merger, holding model, per-image token budget, and decoding fixed. Across Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B — three models spanning PatchMerger and pixel-shuffle designs — pre-merger selection leads post-merger selection on the official TextVQA, DocVQA, and OCRBench splits by +11.0 to +38.4 percentage points and +293 to +432 OCRBench points at 25% retention; paired-bootstrap confidence intervals exclude zero. A pure-stage control at the main-merger boundary retains +30.0 and +12.5 points on TextVQA and DocVQA, while detecting no difference on a  $n = 200$  GQA sample.

We trace the Qwen3-VL gap to selection itself. The merger weakly preserves unit ranks (Spearman  $\rho = 0.14$ – $0.36$ ), preferentially demotes text-stroke units on documents, and a ranking-swap control recovers pre-merger accuracy while keeping exactly the same unit set (Jaccard = 1.000). This motivates **Rank-Before-Merge (RBM)**, a training-free stage change with no learned scorer or query input. RBM is a robust OCR-oriented default, not a universal winner: query-conditioned FastV remains stronger on several reference-grounded and object-centric settings. Four prespecified extensions do not improve over their stronger constituents. At 25% retention, compression remains stage-neutral in speed, increasing throughput by 68% on Qwen3-VL and 1.8–2.5 $\times$  on InternVL3. Code and per-sample evaluation artifacts will be released.

## CCS Concepts

• **Computing methodologies**  $\rightarrow$  **Machine learning**; **Computer vision**; *Natural language processing*.

## Keywords

visual token compression, vision-language models, training-free inference, multimodal foundation models, inference efficiency

## 1 Introduction

**Multimodal inference cost is a visual-token problem.** Vision-language models (VLMs) have become a core class of multimodal foundation models, but an image enters the language model as hundreds to tens of thousands of visual tokens, and the quadratic attention cost and the key-value memory of these tokens dominate multimodal inference. Visual-token compression — pruning or merging the least important tokens before or within the language model — is therefore a standard efficiency lever, with direct consequences for document understanding, scene-text question answering, and OCR, the most token-hungry multimodal workloads. In merger-equipped architectures — the Qwen-VL family, InternVL3, and others that merge  $2\times 2$  patch groups into one token

— the compression debate has run almost entirely over *which* tokens to keep and *how* to score them. The *stage* at which selection happens, before or after the native merger, has not been isolated as an experimental variable. Prior work recognizes that Qwen’s merger already reduces visual-token count [3], and post-encoder merging can degrade OCR-sensitive performance [13, 39]. Yet existing compression work generally treats the native merger as a fixed boundary and does not isolate whether selection should occur immediately before or after it. We isolate that variable and its selection-level mechanism.

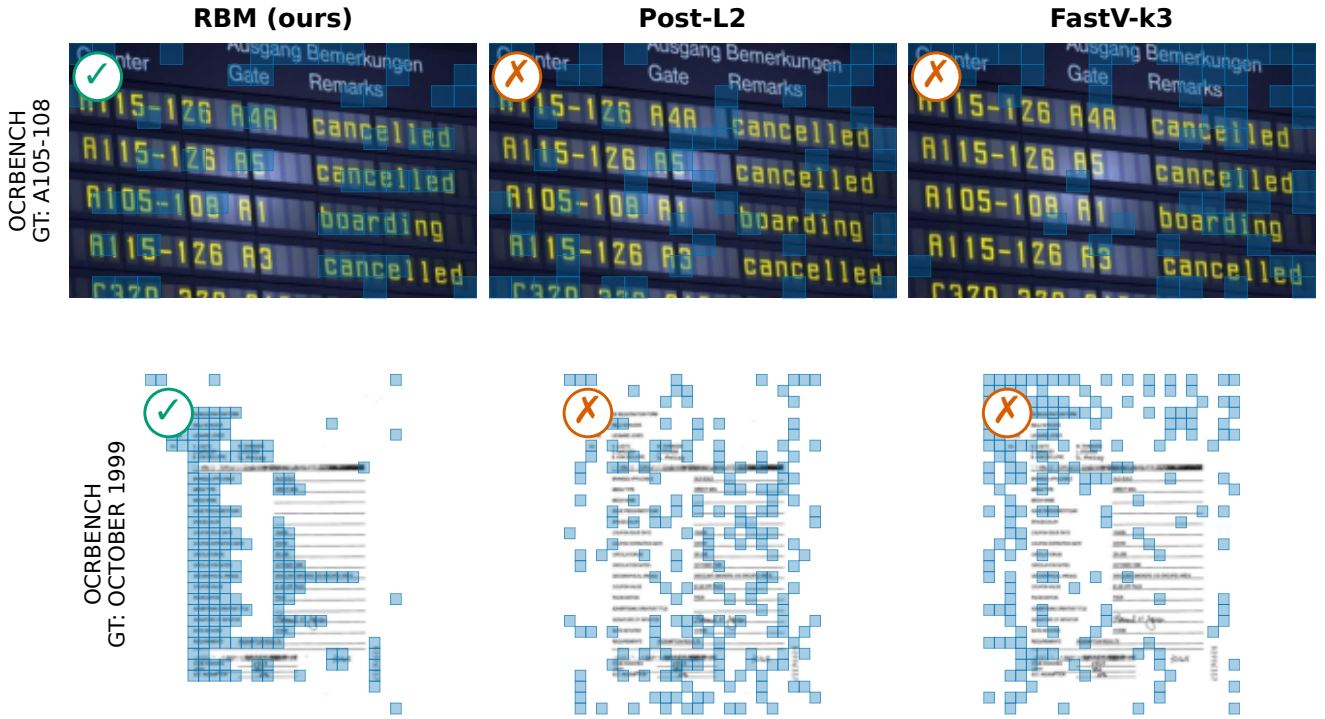
**Causal mechanism.** The learned merger weakly preserves unit ranks (Spearman  $\rho = 0.14$ – $0.36$ ) and, on documents, preferentially demotes high-edge text-stroke units. Running the post-merger path with the pre-merger ranking recovers pre accuracy, while the two paths keep identical unit sets (Jaccard = 1.000 in four model $\times$ benchmark cells). Thus, where tested byte-exactly on Qwen3-VL, the ranking — not the forward path — explains the gap (§5).

**What falls out of it — a stage law across models.** Under iso-model, iso-budget, and iso-selector control, pre-merger selection leads by 11.0–38.4 pp on the official text-dense splits of two Qwen generations and by 34.6–37.4 pp plus 432 OCRBench points on InternVL3, which uses a different merger design. A matched-feature pure-stage control retains the TextVQA/DocVQA effect but detects no GQA difference. The result is therefore workload-conditional, and its gap grows with compression depth (§4).

**What we propose — Rank-Before-Merge (RBM).** Score  $2\times 2$  merge units on their merger-*input* features with a query-blind L2 norm, keep the top- $\kappa$  units, and pass only the survivors through the unmodified native merger. No attention weights, no training, no query input — a pure inference-time stage change. Because the merger is per-unit (a kept unit’s merged token is bit-identical whichever stage selected it), RBM leaves the retained tokens’ bits unchanged and changes only *which* units reach the merger. The method is deliberately minimal, and positioned accordingly: a query-blind, OCR-preserving robust default for text-dense workloads — a *robust default, not uniformly optimal* (§4.4). We freeze it as *plain* RBM and decorate it with nothing, because every decoration we tried is neutral or harmful (§6).

**Contributions.** First, we isolate a selection-level mechanism: rank reshuffling, text-stroke demotion, and a ranking-swap control identify the kept set as the causal lever (§5). Second, we establish a workload-conditioned stage law across three models and two merger designs under iso-selector and iso-budget control (§4). Third, we derive the deliberately minimal RBM rule, map its trade-off against FastV, and show that six prespecified extensions do not improve their stronger constituents (§4.4, §6).

Kept visual units (blue) per selector — two OCR cases



**Figure 1: Three selectors retain different visual evidence.** Two audited OCRBench cases compare RBM, Post-L2, and FastV-k3 at the same  $\kappa = 25\%$  visual-token budget on Qwen3-VL-8B. Blue native-grid cells are the visual units retained by each method; the underlying input remains visible for direct spatial comparison. Vector checks and crosses report answer correctness against the printed ground truth. In both examples, RBM preserves the answer-bearing text and answers correctly, while Post-L2 and FastV-k3 do not. These cases illustrate the selection behavior; aggregate effects and uncertainty are reported in Figure 2.

## 2 Related Work

**Token pruning and merging in VLMs.** Most methods score visual tokens and drop or merge the least important before or within the early LLM layers. **FastV** prunes at the second LLM layer from the attention visual tokens receive [6]; it is our primary same-model, query-conditioned strong baseline — the method that wins its home regime (scene-text/object, low-OCR cells) yet cannot rescue merger-destroyed OCR (§4.4). **PyramidDrop** drops tokens progressively across LLM layers [37]; **FasterVLM** accelerates attention-based pruning [46]; **SparseVLM** explores query-conditioned sparsification [47]; **PruMerge** merges by attention-derived importance [28]; **FitPrune** fits lightweight importance predictors [40]. Other training-free selectors use visual [CLS] attention, graph propagation, or multi-cue suppression [14, 25, 33], while unified recent systems combine importance with diversity or text-guided merging [1, 10, 41]. These methods prune or merge at different encoder-output or LLM-side hook points. None isolates selection immediately before versus after a model’s native spatial merger under iso-model, iso-token, and iso-selector control.

**Token-merging lineage (ToMe → AdaptMerge).** Token merging originates with **ToMe** [4] (bipartite soft matching, post-encoder,

query-blind); **AdaptMerge** [13] adapts it to large multimodal models and reports that language-guided merging closes most of the OCR gap plain ToMe incurs — corroborating evidence for our lossy-merger thesis, cited as support, not as a competitor. This lineage adds a *second*, learned merging stage; we hold the native merger fixed and ask only whether selection should precede or follow it, so it is orthogonal to, and composes with, our stage axis. Learned visual projectors such as TokenPacker condense tokens while preserving fine local cues [21]; unlike our inference-time test, they alter the projector architecture and require model adaptation.

**VisionZip and Qwen-family compression.** **VisionZip** selects “dominant” tokens by CLS-to-patch attention and merges the remainder into “contextual” tokens [39]; a code reading confirms its Qwen path selects *after* the PatchMerger, i.e. on the post-merger side of our axis. Its authors’ own Qwen2.5-VL numbers are a model/stage-mismatched anchor only: OCRBench 81.5 → 70.5 at 50% retention while general tasks hold, with the README caveat that gains are “less striking” because the merger already compresses — the same effect seen from the post-merger side; we claim no head-to-head victory over it (§4.4). Qwen-specific methods **GlimpsePrune** [42] and **VScan** [43] likewise select after the native

merger. Recent layerwise methods instead score token evolution: TransPrune uses token-transition variation [19], while V2Drop exploits variation across adjacent layers [5]. They are complementary to the controlled native-merger stage axis studied here.

**The CVPR 2026 frontier.** Recent work broadens token reduction beyond scalar attention ranks through hierarchical selection [32]. Set coverage, reconstruction error, and functional sensitivity motivate joint or projector-aware selection [9, 15, 17, 26, 38]; learned or strongly conditioned compressors instead model semantic labels, information flow, or head importance [30, 31, 35, 49]. Other work makes token lifetime explicitly layer-dependent [7, 22, 34], or preserves discarded content through object-centric, optimal-transport, or hybrid aggregation [18, 20, 44]. Document- and OCR-specific studies further show that generic accuracy can hide lost textual evidence [8, 23]. These advances reinforce the need to localize the intervention stage, but none performs our matched pre/post test at a frozen model’s native spatial merger.

**The design point we isolate.** We do not claim a topological vacuum. QuietPrune-style in-ViT pruning [11] is a different, earlier, trained stage; FastV/PyramidDrop select inside the LLM; VisionZip selects on the opposite (post-merger) side. The combination we isolate is narrower: on a merger-equipped VLM, leaving the native merger untouched, scoring the merger-*input*  $2 \times 2$  units by a query-blind saliency norm, keeping the top- $\kappa$ , and handing only the survivors to the unmodified native merger — together with the causal mechanism (M1–M3) explaining *why* this stage dominates on text-dense workloads, which the methods above neither report nor control for.

**Efficiency evaluation of VLMs.** Reported speedups depend heavily on engine and protocol. We adopt the lmms-eval family of *official* scorers for accuracy [45]; EffiVLM-Bench independently emphasizes cross-backbone, multi-metric acceleration evaluation [36]. We confine all throughput claims to a single engine (§7).

## 3 Method — Rank-Before-Merge (RBM)

### 3.1 Models, mergers, and the stage axis

We study three merger-equipped VLMs with two merger designs. Qwen3-VL-8B and Qwen2.5-VL-7B use a learned PatchMerger [2, 3]; InternVL3-8B uses pixel-shuffle downsampling [48]. All are served in bf16, eager attention, vLLM 0.19 (V1) [16], single A40 (46 GB), greedy decoding. The shared pipeline is: image  $\rightarrow$  ViT patch tokens  $\rightarrow$  native merger ( $2 \times 2$  groups  $\rightarrow$  one token)  $\rightarrow$  language model. The Qwen merger is *not* an averaging pool: the four patch features of each  $2 \times 2$  group are concatenated and projected by a small learned MLP (LayerNorm  $\rightarrow$  fc1  $\rightarrow$  GELU  $\rightarrow$  fc2) to one unit vector; the lossy step is the four-to-one nonlinear projection, so a kept unit’s representation is a nonlinear function of its four patches. We verified this against the served implementation on both Qwen generations.

**The stage axis (the experimental variable).** Two hook points differ only in *where* selection happens, with model, token count, and scorer held fixed:

- **Pre-merger (RBM).** Score all  $N \times 2$  units on their *merger-input* features, keep the top- $\kappa N$ , and pass only the survivors through the unmodified native merger (and, for Qwen3-VL,

each deepstack merger). The merger operates exactly as in the uncompressed model, on a subset of units.

- **Post stage (contrast).** Either score the  $N$  *merged* units on *merger-output* features and keep the top- $\kappa N$  (the stage used by published compression methods for merger-equipped VLMs, including VisionZip’s Qwen path), or — the stronger, query-conditioned contrast — prune inside the LLM by layer-2 attention (the FastV stage). The main matrix (Table 1) uses the post-merger-L2 contrast, iso-selector with the pre arm; the layer-2 FastV contrast is used only in Table 2.

**Budget definition.** Retention  $\kappa$  is defined over *merge units* relative to each image’s **own** full unit count, so two methods at the same  $\kappa$  feed the language model the same number of visual tokens per image; we verify equality of the mean post-merger token count per benchmark (**iso-token** control), so any accuracy difference is attributable to *which* units survive and the representational state on which selection is made, not to token count. We report  $\kappa \in \{0.25, 0.125\}$  (25% / 12.5% retention); per-benchmark token counts are printed with each table.

### 3.2 The L2 selector as a control variable

To isolate the *stage*, we deliberately use the simplest possible **text-agnostic, query-blind** scorer, identical at both hook points: the L2 norm of the unit feature vector,  $s(u) = \|f_u\|_2$ , computed on merger-input features for pre-merger selection and on merger-output features for post-merger selection. A strong task- or query-aware scorer would confound *scoring quality* with *stage*; with an identical, saliency-free scorer, the only manipulated variable is the hook point. We freeze this as **plain RBM** and add no variant. §5 additionally reports a mechanism-side probe: a *variance-weighted* variant of the same query-blind scorer,  $s_{\text{freq}}(u) = \alpha z(\|f_u\|_2) + \beta z(\text{Var}_u)$  (both terms per-image standardized, so the ratio  $\alpha:\beta$  is a genuine trade-off knob rather than an arbitrary scale), improves text-dense accuracy under aggressive compression (+1.2 pp TextVQA at 25% retention, +0.5 pp at 50%) yet is not a general upgrade: it regresses −1.6 to −2.2 pp on DocVQA, OCR-Bench, and GQA at 25%. We therefore treat it as corroborating evidence for §5.2’s text-stroke hypothesis rather than as a decoration of the frozen selector (§6e). §5.4 reports that the law survives replacing L2 with a second scorer family (global-centroid attention) on Qwen3-VL, indicating the effect is not an L2 artifact. Unlike the merging lineage of §2, RBM introduces no second learned merger and no query input: it manipulates only the *set* of units the native merger sees — which is exactly what makes the pre-vs-post contrast a clean stage experiment rather than a scorer comparison.

### 3.3 Rank-Before-Merge

Given an image with  $N$  merge units, compute  $s(u)$  on merger-input features for every unit, retain the top  $k = \kappa N$ , and invoke the native merger(s) on the retained units only. Unit identity is shared across the main and deepstack streams (deepstack mergers process intermediate-depth features of the *same* units), so pruning a unit removes it from every stream. No attention weights are required and no parameters are trained.



**Algorithm 1** Rank-Before-Merge (RBM), one image

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**Require:** image  $x$ ; retention  $\kappa$ ; query-blind scorer  $s = \|\cdot\|_2$

**Ensure:** compressed visual token sequence  $z$

- 1:  $P \leftarrow$  ViT patch tokens of  $x$  {16 px footprint}
- 2:  $U \leftarrow$  group  $P$  into  $N$  native  $2 \times 2$  units {32 px footprint}
- 3: **for** each unit  $u \in U$  **do**
- 4:    $\text{score}[u] \leftarrow \|f_{\text{in}}(u)\|_2$  {merger-input features}
- 5: **end for**
- 6:  $K \leftarrow$  top- $k$  units by score,  $k = \text{round}(\kappa \cdot N)$  {per-image budget}
- 7:  $z_{\text{main}} \leftarrow \text{NativeMerger}(\{u : u \in K\})$  {merge survivors only}
- 8:  $z_{\text{ds}} \leftarrow \text{DeepstackMergers}(\{u : u \in K\})$  {Qwen3-VL only}
- 9:  $z \leftarrow \text{concat}(z_{\text{main}}, z_{\text{ds}})$
- 10: **return**  $z$  {feed standard LLM}

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**Unit equivalence (the structural invariant).** Because the mask is at *unit* granularity and the merger is per-unit, a kept unit's merged token is **bit-identical** regardless of stage — the merger sees all four patches of a kept unit either way. Selection cannot prevent intra-unit combination; it changes only *which* units reach the merger. Pre-merger selection keeps units on their raw (merger-input) saliency; post-stage selection keeps them on a saliency the merger has already rewritten (§5). This invariant is what makes the ranking-swap control of §5.3 decisive.

**Post-merger selection (the contrast):** replace lines 3–5 of Algorithm 1 by  $U' \leftarrow \text{NativeMerger}(U)$ ;  $\text{score}'[u'] \leftarrow \|f_{\text{out}}(u')\|_2$  on merger-output features;  $K' \leftarrow$  top- $k$  by score'; keep( $K'$ ). Everything downstream is identical.

**Configuration details.** Per-family tap points (Qwen3-VL deepstack[0], Qwen2.5-VL main merger, InternVL3 pixel-shuffle input) and the Qwen2.5-VL position-cursor fix are detailed in Supplementary § S8, including the cross-generation configuration diagnostic.

## 4 Main Experiments

### 4.1 Setup

**Benchmarks and metrics.** We use TextVQA VQA accuracy [29], DocVQA ANLS [27], the official OCR-Bench score [24], and GQA normalized exact match [12]. Official split sizes are 5000, 5349, 1000, and 12578, respectively. Scorers are verbatim ports of the lmms-eval family [45], with ground-truth self-test passing 200/200.

**Engines and fairness protocol.** RBM and the none / post cells run in vLLM; the FastV baseline runs in an independent HuggingFace transformers eager harness (it has no vLLM path). Accuracy is comparable across engines (the harness bit-degrades to native at zero drop, 8/8 per-sample identical; HF-vs-vLLM none cells agree 16/16); **throughput is not** — all efficiency numbers are confined to vLLM (§7). Fairness is enforced by the same keep ratio relative to each image's own token count and by reporting mean post-merger token counts.

**Significance.** Each pre-vs-post  $\Delta$  pairs the same samples under greedy decoding. Primary inference uses a 20,000-resample paired bootstrap confidence interval on per-item metric differences, with a paired sign-flip permutation test. McNemar is reported additionally for binary OCRBench containment and GQA exact match. This

preserves partial credit in VQA-accuracy and ANLS rather than binarizing those metrics.

### 4.2 Main result across three models — official full splits (Table 1)

**The stage law holds on both generations.** Every text-dense  $\Delta_{25}$  is large and significant: RBM leads post-merger selection by +38.4 pp (TextVQA), +24.3 pp (DocVQA), +363 pts (OCR-Bench; 547 vs 184, 3.0 $\times$ ) on Qwen3-VL, and by +26.1 pp, +11.0 pp, +293 pts (476 vs 183, 2.6 $\times$ ) on Qwen2.5-VL; all paired-bootstrap 95% CIs exclude zero. The gap is robust to deeper compression: at  $\kappa = 0.125$  the post stage degrades much faster (Qwen3-VL OCR-Bench 184  $\rightarrow$  53 vs RBM 547  $\rightarrow$  350, widening the pre/post ratio from 3.0 $\times$  to 6.6 $\times$ ), and all six text-dense  $\Delta_{12.5}$  remain significant under paired metric-level inference.

**A pure-stage confound control (pre-final).** The original pre arm on Qwen3-VL ranks at the deepstack[0]-input tap (layer 8, far upstream) and slices all four mergers, while the post arm ranks at the main-merger *output* — so the pre-vs-post contrast differs in *both* stage and feature space. To isolate the stage alone we run a **pre-final** control that ranks L2 at the main-merger's *own input* (final ViT-block features, the same feature space the post arm ranks *after* the merger) and selects the top- $\kappa$  merge-units from the visual output, leaving the deepstack mergers untouched ( $n = 200$ , Qwen3-VL, greedy,  $\kappa = 0.25$ , L2, iso-config with Table 1; iso-token verified, 200/200 per benchmark). The only difference between pre-final and post is then *before versus after* the lossy  $2 \times 2$  main merger — a pure stage variable.

Pre-final still beats post by +30.0 pp on TextVQA and +12.5 pp on DocVQA — the same direction and order as the original pre-vs-post gap (+38.3/+26.5 pp at  $n = 200$ ). The full control table and paired uncertainty are in Supplementary Table S7b. The text-dense pre $\gg$ post claim is therefore a **true stage effect** (ranking before the lossy merger preserves text), *not* a deepstack[0]-input versus main-merger-output feature-space artifact; the pure-stage control is valid (iso-token verified).

**Object-centric GQA — no detected pure-stage difference.** On GQA, pre-final *equals* post (0.450 = 0.450, 0.0 pp, paired mean  $0.000 \pm 0.032$ ): this experiment detects no difference between ranking before and after the merger on object-QA, consistent with the mechanism (the merger does not preferentially damage object features; M2  $\rho = 0.04$  on GQA, §5). The original pre-vs-post contrast on Qwen GQA (−2.8 pp Qwen3-VL / −2.6 pp Qwen2.5-VL at full  $n = 12578$ ; primary McNemar  $z = -8.0/-5.7$ ; independent-binomial  $z = 4.5/4.1$ ; 83.9% / 83.8% per-question agreement; the official exact-match rescore gives a concordant 8.1/7.1) is therefore *not* a true stage reversal: it is an artifact of the deepstack[0]-input (layer-8) ranking being poorly suited to object-QA, dissolved once the ranking taps the main-merger input (pre-final, 0.0 pp). Its magnitude is an order of magnitude below any text-dense  $\Delta$  (1/10–1/14 of them), and **no text-dense benchmark shows any crossover at the native-resolution full splits**. We therefore claim **no detected pure-stage difference** on this object-centric sample; this is not an equivalence test. All other sections refer here for the GQA statistics.

**A no-merger negative control (LLaVA-1.5).** LLaVA-1.5-7B has a per-token MLP projector (Linear–GELU–Linear, no  $2 \times 2$  spatial

**Table 1: Official full-split metrics ( $n = 5000/5349/1000/12578$ ), greedy, across Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B. RBM = pre-merger L2 (ours); post = post-merger L2 at the same token budget.  $\Delta_{25}$  = pre – post at  $\kappa = 0.25$ . † marks the only significant post-stage leads (GQA); these use the deepstack[0]-input (layer-8) pre ranking and are reframed as a layer-8 ranking artifact by the pure-stage control in §4.2 (pre-final gives 0.0 pp on GQA). InternVL3 uses a different merger and family-specific imaging configuration; compare within-model deltas, not cross-model absolute values. Full protocol and skip accounting are in the supplementary material.**

| Model         | Benchmark (metric) | none   | RBM @25%      | post @25% | $\Delta_{25}$   | RBM @12.5% | post @12.5% |
|---------------|--------------------|--------|---------------|-----------|-----------------|------------|-------------|
| Qwen3-VL-8B   | TextVQA (VQA-acc)  | 0.844  | <b>0.605</b>  | 0.222     | <b>+38.4 pp</b> | 0.472      | 0.132       |
|               | DocVQA (ANLS)      | 0.956  | <b>0.481</b>  | 0.238     | <b>+24.3 pp</b> | 0.352      | 0.103       |
|               | OCR-Bench (/1000)  | 760    | <b>547</b>    | 184       | <b>+363 pts</b> | 350        | 53          |
|               | GQA (exact match)  | 0.616  | 0.449         | 0.477     | –2.8 pp †       | —          | —           |
| Qwen2.5-VL-7B | TextVQA (VQA-acc)  | 0.862  | <b>0.702</b>  | 0.442     | <b>+26.1 pp</b> | 0.597      | 0.319       |
|               | DocVQA (ANLS)      | 0.949  | <b>0.636</b>  | 0.526     | <b>+11.0 pp</b> | 0.455      | 0.245       |
|               | OCR-Bench (/1000)  | 817    | <b>476</b>    | 183       | <b>+293 pts</b> | 335        | 67          |
|               | GQA (exact match)  | 0.604  | 0.559         | 0.585     | –2.6 pp †       | —          | —           |
| InternVL3-8B  | TextVQA (VQA-acc)  | 0.8338 | <b>0.7890</b> | 0.4148    | <b>+37.4 pp</b> | 0.7230     | 0.3064      |
|               | DocVQA (ANLS)      | 0.9221 | <b>0.7284</b> | 0.3820    | <b>+34.6 pp</b> | 0.5054     | 0.2451      |
|               | OCR-Bench (/1000)  | 852    | <b>753</b>    | 321       | <b>+432 pts</b> | —          | —           |
|               | GQA (exact match)  | 0.6293 | 0.5993        | 0.6031    | –0.4 pp (n.s.)  | —          | —           |

**Table notes.** (a) Qwen3-VL none DocVQA was re-run at max-model-len 49,152 and completed with 0/5349 skips (ANLS 0.956); the pre/post cells it anchors ran at the same native-resolution configuration and are directly comparable to each other. (b) OCR-Bench none cells skip 18 (Qwen3-VL) / 24 (Qwen2.5-VL) of 1000 long-OCR images on context overrun, scored 0 (conservative; compressed cells skip  $\leq 5$ ), so the uncompressed anchor is understated and the relative gaps are, if anything, conservative. (c) GQA was evaluated at  $\kappa = 0.25$  only on the full split. (d) Iso-token control: at each model  $\times$  benchmark, pre and post have identical mean post-merger token counts, with one exception – the Qwen2.5-VL OCR-Bench @25% RBM cell used a 4 M-pixel cap (mean tokens 229.6 vs post 282.0), so its +293-pt gap is conservative. (e) Primary inference uses paired-bootstrap 95% CIs and paired sign-flip permutation tests on the official per-item metric. All Qwen text-dense intervals exclude zero; Figure 2 shows the intervals. McNemar is additional for binary OCRBench/GQA outcomes. (f) InternVL3 uses pixel-shuffle downsampling and a 4 M-pixel DocVQA cap; its pre/post prompt token counts are identical within every cell. Its GQA –0.4 pp difference is indistinguishable: the paired 95% CI [–0.93, +0.17] crosses 0 (permutation  $p = 0.18$ ), so we claim no GQA stage lead on InternVL3; no preregistered equivalence bound exists, so we do not call it a statistical tie.

averaging). Running the same pre-vs-post L2 protocol (pre-MLP vs post-MLP ranking, keep 25%,  $n=200$ , greedy) yields *no* significant stage effect: TextVQA  $\Delta = -2.17$  pp (McNemar  $p=0.125$ ; none 0.452, pre 0.087, post 0.108), GQA  $\Delta = -0.5$  pp ( $p=1.0$ ; none 0.575, pre 0.375, post 0.380) – versus the merger-equipped Qwen3-VL’s +38.4 pp ( $p \approx 5 \times 10^{-5}$ ) on TextVQA. In this no-merger control the trend is slightly reversed (post  $\geq$  pre, n.s.). Together with the merger-equipped results, this is consistent with the spatial merger being a key condition for the large text-dense pre  $>$  post gap. A single negative-control architecture cannot establish necessity or an if-and-only-if law.

### 4.3 Generalization to a third family – InternVL3-8B (Table 1)

To test whether the stage law is a Qwen-family artifact or a property of merger-equipped VLMs, we run RBM and the post stage on **InternVL3-8B**, whose merger is pixel-shuffle downsampling – a different design from Qwen’s PatchMerger. The post arm here is **post-merger L2 top-k** – the *same* query-blind L2 selector as the pre arm, at an identical token budget (ptid equal across arms), so the InternVL3 block of Table 1 is a **pure stage-variable experiment under iso-selector / iso-budget control** at 25% retention. The query-conditioned in-layer FastV contrast is a separate campaign, reported in Table 2.

**Configuration boundary.** InternVL3 DocVQA uses a 4 M-pixel cap, while the Qwen same-scope baseline cells of §4.4 use 600k; cross-model absolute values are not comparable. At 25% retention

RBM costs –19.4 pp vs none on InternVL3 DocVQA and –99/1000 on OCR-Bench, versus only –4.5 pp on TextVQA and –3.0 pp on GQA. The claim is therefore the large within-model pre  $>$  post margin, not losslessness on every text-dense workload.

At 12.5% retention ( $r = 0.875$ ), TextVQA is 0.7230 pre vs post 0.3064 (**+41.7 pp**); DocVQA pre 0.5054 vs post 0.2451 (**+26.0 pp**) – the gap *widens* under deeper compression, as on Qwen.

**Cross-family isomorphism.** InternVL3’s pre–post text-dense gaps (+37.4/+34.6 pp / +432 pts) are the same order and direction as Qwen3-VL’s (+38.4/+24.3/+363 pts) on all three text-dense benchmarks, and GQA stays within the red line – indistinguishable here (–0.4 pp) versus Qwen3-VL’s apparent post edge (+2.8 pp, a deepstack[0]-input layer-8 ranking artifact dissolved by the pure-stage control, §4.2), with no text-dense crossover at either family’s native imaging configuration. The stage law – the same query-blind L2 selector leads when applied before the merger because selection deferred past it loses raw-patch text information to the merger’s ranking rewrite (§5), at identical selector and budget – is therefore established on a **third model family with a different merger**, supporting the “merger-equipped VLMs” generalization rather than a family-specific claim.

**The ranking-swap identity generalizes to InternVL3.** The M3 control (§5) – run the post-merger forward path but select units with the pre-merger ranking – recovers pre-merger accuracy exactly on InternVL3 too: swap vs pre is +0.00 pp (TextVQA), +0.11 pp (DocVQA), –0.50 pp (GQA), with kept-set Jaccard(pre,swap)= 1.0 and per-sample answer-identity 0.98–1.0 ( $n = 200$ , iso-token, no

swap fallback). The pre>post gap is therefore a pure ranking effect — which units are kept, not the forward path — on a *second* merger architecture (pixel-shuffle, not just Qwen’s PatchMerger), extending the M3 evidence to a second merger design.

#### 4.4 A workload-conditioned regime map — RBM vs the query-conditioned strong baseline FastV (Table 2)

FastV is the query-conditioned strong baseline in this comparison — it selects inside the LLM after cross-attention has mixed in the question — and we give it every advantage: we compare RBM against FastV at its **best layer**  $K$  ( $K = 3$ ; a  $K$ -sensitivity probe found  $K = 3$  best), reported at identical  $n$ /scope with paired skips and official rescoring, so the comparison cannot be attributed to a handicapped  $K$  or a mismatched sample set. We read the result as a **workload-conditioned regime map** — where query-conditioned selection wins, where it fails, and why — not as a single-winner race. At its canonical progressive schedule PyramidDrop is near-lossless but occupies a different budget point (equivalent mean retention  $0.625 \approx 2.1 \times$  ours); forced to an honest iso-25% schedule it collapses (TextVQA 0.073 / GQA 0.305 / OCR 0.005 on Qwen3-VL), whereas RBM holds (0.60 / 0.42 / 0.58). There is therefore no universal winner.

**Workload-conditioned regime map. FastV wins its home regime.** Query-conditioned FastV-k3 leads RBM on all three Qwen3-VL text-dense benchmarks (+17.2/+8.9/+16.2 pp) and on Qwen2.5-VL TextVQA (+7.8 pp) — the reference-grounded, low-OCR workloads. This is the query-dependent headroom made reachable: *which* region matters for *this question* becomes visible only once cross-attention has seen the question, exactly where FastV operates and exactly where the prespecified query-level gate of §6 could not reach before the merger. **FastV fails the OCR regime.** RBM in turn wins OCR-Bench on **both** families (+16.0 pp Qwen3-VL, +8.5 pp Qwen2.5-VL); the Qwen2.5-VL  $n = 200$  GQA/DocVQA cells favor RBM by +4.5/+2.1 pp but their paired 95% CIs cross 0 ( $[-10.5, +1.5]$  /  $[-10.1, +5.8]$ ), so they are exploratory/inconclusive at  $n = 200$ , not statistically robust RBM edges. The OCR split is structural, not a tuning artifact: FastV’s collapse survives its own best  $K$  — dense OCR information is destroyed once text patches are discarded inside the LLM, and no query conditioning recovers information already thrown away; only pre-merger selection, which keeps raw patches, preserves it. **The map is the two failure modes of §5.4.** Documents = selection misdirection — the merger’s reshuffle misleads selection made after it away from text, so the OCR regime is lost by the selectors that tap the stream post-merger; scene text = feature degradation under a budget every 25%-keep method pays, and object scenes = near-null reshuffle — together the low-OCR regime where the query-dependent headroom lives. We report FastV’s wins honestly: FastV is a strong query-conditioned baseline that wins its regime; RBM is the robust default (never collapses, uniquely holds dense OCR), not a method that leads query-conditioned pruning everywhere.

#### 5 Mechanism — Why the Merger Is Lossy

We give the pre-vs-post gap a mechanistic account in three parts (M1–M3), measured on **Qwen3-VL-8B** (M1/M2 on a deterministic

**Table 2: Workload-conditioned regime map — same-scope paired comparison of FastV at best  $K = 3$  (the query-conditioned strong baseline) vs RBM (pre, the query-blind robust default) @25% retention, both Qwen models, official rescoring. Cells are paired within each row (identical  $n$ , identical attempted samples; skip parity in note ii). Margin = winner’s absolute lead; the winner column is the regime read: FastV’s wins mark its reference-grounded home regime, RBM’s wins mark the OCR regime (§5.4).**

| Benchmark (scope; pixel cap) | $n$   | FastV-k3      | RBM @25%     | $\Delta$ (F/R)      |
|------------------------------|-------|---------------|--------------|---------------------|
| <b>Qwen3-VL-8B</b>           |       |               |              |                     |
| TextVQA (full; default)      | 5000  | <b>0.7771</b> | 0.605        | <b>+17.2 F</b>      |
| GQA (full; default)          | 12578 | <b>0.5376</b> | 0.449        | <b>+8.9 F</b>       |
| DocVQA (dev; 600k cap)       | 200   | <b>0.5863</b> | 0.4239       | <b>+16.2 F</b>      |
| OCR-Bench (dev; native res.) | 200   | 0.415         | <b>0.575</b> | <b>+16.0 R</b>      |
| <b>Qwen2.5-VL-7B</b>         |       |               |              |                     |
| TextVQA (dev; default)       | 200   | <b>0.7467</b> | 0.6683       | <b>+7.8 F</b>       |
| GQA (dev; default)           | 200   | 0.475         | 0.520        | +4.5 R <sup>‡</sup> |
| DocVQA (dev; 600k cap)       | 200   | 0.4852        | 0.5062       | +2.1 R <sup>‡</sup> |
| OCR-Bench (dev; native res.) | 200   | 0.285         | <b>0.370</b> | <b>+8.5 R</b>       |

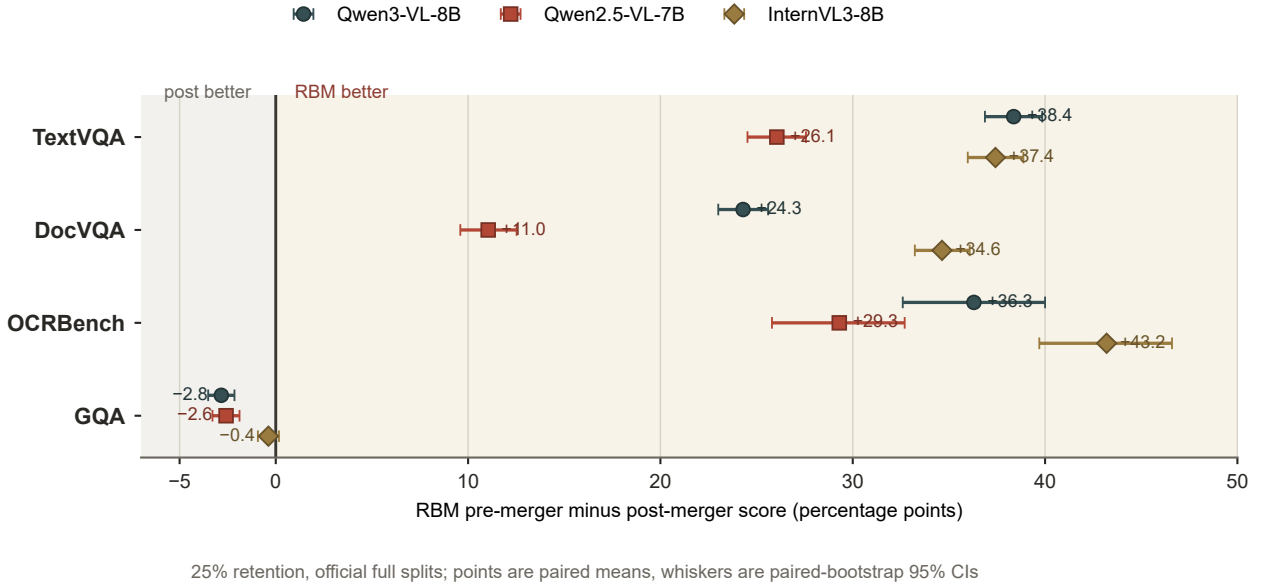
**Table notes.** F = FastV; R = RBM;  $\Delta$  is the winner’s absolute lead in percentage points. (i) Scope and pixels:  $n$  is printed per row; DocVQA cells run at a 600k pixel cap on both arms (iso-pixel within this table); OCR-Bench runs at native resolution, which is what triggers the skips (note ii); TextVQA/GQA use each model’s default imaging configuration. (ii) OCR-Bench skips: native-resolution giant OCR images overrun the context on  $\approx 10\%$  of the  $n = 200$  set (19/200 Qwen3-VL, 20/200 Qwen2.5-VL); the *identical* hard samples are skipped under both arms (same paired batch, mode-independent) and scores are over attempted samples — conservative and symmetric. All other cells skip  $\leq 5$  (Qwen3-VL TextVQA FastV, 5/5000) or 0. (iii) Harness: every  $n = 200$  cell on both sides comes from the same HF transformers eager-attention harness with family-correct native positions; the Qwen3-VL TextVQA/GQA RBM references are the vLLM full-split cells of Table 1, for which engine parity is validated (§4.1; HF-pre vs vLLM-pre @25% further agree 14/16 with kept-set Jaccard 0.986, §6d). An early Qwen2.5-VL  $\times$  pre position-harness degeneration (vllm-mimic layout) was detected and replaced by the native layout before any reported number was produced; no reported cell comes from the degenerate path (on Qwen3-VL, native vs mimic differ, 0.575 vs 0.525, same ordering). (iv) Budget: FastV-k3  $\approx 214$  vs RBM  $\approx 213$  mean visual tokens/request (Qwen3-VL TextVQA, 25% retention; the  $n = 200$  probe measured 208) — approximately matched, stated per cell; this is “each method at its strong configuration,” not a byte-iso-budget cell. (v) <sup>‡</sup>The paired 95% CI crosses 0 at  $n = 200$  (GQA  $[-10.5, +1.5]$ , DocVQA  $[-10.1, +5.8]$ ); these two cells are exploratory/inconclusive, not statistically robust RBM edges.

seed-0  $n = 64$  sample per benchmark; M3 on  $n = 200$  with the same invocation as the headline cells), then state its architectural scope.

#### 5.1 M1 — the merger reshuffles unit ranks

Per image we correlate the *pre* ranking (merger-input unit L2) with the *post* ranking (merged-token L2) over all units. The two rankings are barely related, and the reshuffle tracks text density: Spearman  $\rho = 0.137$  (DocVQA, text-dense document)  $< 0.332$  (TextVQA, text-dense scene)  $< 0.360$  (GQA, object-centric), with the decision-relevant top-25% overlap rising in the same order (Jaccard@25% = 0.180  $< 0.243 < 0.278$ ; full table in Supplementary Table S1). The merger’s nonlinear projection thus reorders unit saliency almost from scratch ( $\rho = 0.14$ – $0.36$ , nowhere near 1); both the full-ranking correlation and the decision-relevant top-25%





**Figure 2: The text-dense stage effect replicates across models and merger designs. Points are paired pre–post metric differences at 25% retention; whiskers are paired-bootstrap 95% confidence intervals on the official full splits. OCRBench’s /1000 score is divided by 10 for a common percentage-point axis. Qwen uses PatchMerger; InternVL3 uses pixel shuffle. GQA bounds the result: Qwen’s original layer-8 pre-ranking favors post, while InternVL3’s interval crosses zero. Values correspond to Table 1.**

overlap rise from the most text-dense benchmark to the object-centric one, so the reshuffle is largest on DocVQA and smallest on GQA – yet substantial on every benchmark.

## 5.2 M2 – the demoted units are text-stroke units

With  $\text{rank\_shift} = \text{post\_rank} - \text{pre\_rank}$  ( $+$   $\Rightarrow$  demoted) and per-unit Sobel edge energy as a text-stroke proxy, the merger preferentially *demotes* high-edge units at  $\kappa = 25\%$ :  $\rho(\text{rank\_shift}, \text{edge}) = +0.439$  on DocVQA,  $+0.155$  on TextVQA,  $+0.036$  on GQA –  $\sim 12\times$  stronger on documents than on object scenes. The units post drops that pre keeps are the highest-edge of all: on DocVQA, mean Sobel **0.641 vs 0.124** for the reverse group, with **92% above the per-image median (vs 35%)** – post-merger selection is **systematically anti-text on documents**. (The *direction*, we hypothesize, is set by the merger’s training on natural content, which attenuates high-frequency stroke energy; this is a testable hypothesis, not a measured result.) The full M2 table (per-benchmark Sobel means and median fractions) is Supplementary Table S2.

## 5.3 M3 – a ranking-swap causal control, and the decisive kept-set identity

By unit equivalence (§3.3), holding the *forward path* fixed at post and swapping in the *pre ranking* must reproduce pre accuracy if the gap is purely a ranking effect. It does:

Swapping in the pre ranking recovers full pre-merger accuracy (exactly on DocVQA; within 2/200 greedy-decode answers on TextVQA,  $\epsilon$ -level kernel numerics confirmed by a 200/200 byte-identical independent rerun; full values are in Supplementary S9)

and erases the collapse ( $+38.3$  pp TextVQA,  $+26.5$  pp DocVQA over post). The swap path agrees with post on only 40/200 (TextVQA) and 63/200 (DocVQA) answers – without the pre ranking, the same forward path produces the collapse.

**The decisive hardening – kept-set identity on both architectures.** Emitting the kept unit indices from both paths and comparing directly: **Jaccard(swap-kept, pre-top- $\kappa$ ) = 1.000 in all four cells** (Qwen3-VL and Qwen2.5-VL  $\times$  DocVQA and TextVQA;  $n = 32$ , chance level  $\kappa/N = 0.25$ ). The two paths select *exactly the same units*. The small swap-vs-pre answer residual on Qwen2.5-VL ( $+6.7$  pp DocVQA at  $n = 32$ ) traces to an implementation confound of the swap control – window-attention *reverse\_indices* (three recovery sites on Qwen2.5-VL, zero on Qwen3-VL) map identical index sets to physical units in different orderings – while batch-dependent merging is structurally excluded (the merger is per-token LN+MLP). **Verdict: on Qwen3-VL, the tested pre>post gap is attributable to the ranking alone at selection level** (byte-identical); Qwen2.5-VL corroborates kept-set identity but retains an ordering-confound residual. The merged representations of kept units carry no stage-dependent loss on Qwen3-VL (byte-exact); on Qwen2.5-VL the kept *set* is identical but the ordering confound is not eliminated, so we scope the no-loss statement to the byte-exact family.

## 5.4 Failure modes and scope boundaries

DocVQA exhibits selection misdirection: post ranking moves away from text and the swap fully restores accuracy. TextVQA combines rank corruption with a budget effect, while GQA has near-null text

directionality ( $\rho = 0.04$ ). The stage effect is therefore workload-conditional. It retains sign and magnitude under a global-centroid attention scorer on Qwen3-VL, but that proxy degrades on Qwen2.5-VL; L2 remains the paper's fixed selector. The ranking law replicates on Qwen2.5-VL and at the accuracy level on InternVL3 (Table 1), whereas the causal decomposition is claimed only for byte-exact Qwen3-VL. A low 600k pixel cap reverses shallow Qwen2.5-VL DocVQA; the native-resolution full split remains the headline configuration (§8).

**Retention curve.** The stages are nearly indistinguishable at 75% retention; their gap then widens monotonically through 25% on both Qwen models (full values in Supplementary Table S3,  $n = 200$ ).

**Merger mechanism.** Figure 3 connects the stage effect to the native merger. Across four workloads, pre- and post-merger rankings share only moderate rank correlation and top-quartile overlap. Units rich in text-stroke edge energy are preferentially demoted after merging. Finally, a Qwen3-VL ranking-swap control keeps the selected unit set fixed (Jaccard = 1.000) while restoring the pre-merger ordering, separating ranking damage from a change in token identity.

## 6 Negative Results — Tested Extensions Do Not Compose

We prespecified six query-aware, regime-aware, or composite extensions and report all six as failures on Qwen3-VL-8B. **(a) Image-level routing:** pre is at least as accurate as post on 84–97% of images per benchmark; the best router scores below always-pre (0.484 vs 0.494) and far below the per-sample oracle (0.576). On pooled  $N = 774$ , only ~27% of the oracle gap is workload-level, leaving ~73% sample- and query-dependent. **(b) Merger-aware masking:** its selected  $t = 0.5$  hybrid gains 2.3 pp on TextVQA but loses 8.0 pp on OCR-Bench and gains nothing on GQA; no candidate passes the no-OCR-regression plus GQA-gain gate. **(c) Query-embedding blending:** on a disjoint development slice every  $\lambda > 0$  underperforms plain L2 (−1.7/−5.2/−3.3 pp for  $\lambda = 0.3/0.5/0.7$ ), so the prespecified rule selects  $\lambda = 0$ . **(d) RBM→FastV cascade:** for  $n = 200 \times 4$  benchmark samples and two budgets, it misses the 0.5-pp Pareto margin in all eight cells, typically by 8–18 pp, and never significantly beats both arms. Complete gates, paired counts, and harness controls are in Supplementary S4b/S9.3. **(e) A variance-weighted (frequency-aware) selector:** the mechanism-side probe of §3.2 ( $\alpha=1$ ,  $\beta=0.6$ , selected in two grid rounds on TextVQA-500 at 25% retention) improves the text-dense cell (+1.2 pp TextVQA@25%, +0.5 pp at 50%) but regresses −1.6 to −2.2 pp on DocVQA@25%, OCR-Bench@25%, and GQA@25%; no candidate passes a no-regression gate across the four benchmarks, so we freeze the L2 selector and treat the improvement as hypothesis support only. **(f) An adaptive PRE/POST stage router:** a query-blind workload detector on merger-input features (high-frequency tail ratio + L2-entropy) routes each image to the stage predicted to be stronger. Tuned on TextVQA it reduces to always-PRE — statistically indistinguishable from plain RBM on every benchmark (TextVQA/DocVQA/OCR-Bench regression = 0.0 pp, but GQA gain = +0.0 pp, below the required +1 pp); raising the threshold to route scene images to POST collapses TextVQA by a further 7 pp. The GQA and TextVQA workload signatures

overlap at the unit level, so no single global threshold reconciles the two acceptance criteria.

These results bound, rather than close, the tested design space: image-only signals cannot access the query-dependent oracle headroom, while cascading discards evidence needed by the second selector. We therefore freeze plain RBM as the robust text-dense/OCR default and select a FastV-class method for scene-text/object-centric traffic; within the tested configurations, runtime routing and cascading offer no benefit.

## 7 Efficiency — Supporting Evidence

Efficiency is supporting evidence, measured only in vLLM 0.19 V1 offline on one A40. **Stage-neutral:** across 75/50/25% retention, pre vs post throughput is 4.61/4.52, 5.39/5.54, and 6.91/6.89 req/s; five-repeat measurements remain within 3% with no consistent direction. **Compression scaling:** relative to no compression, throughput rises 12/31/68% and 25%-retention latency falls from 0.36 to 0.23 s. InternVL3 independently yields 1.8–2.5× throughput at 25% retention, again with pre  $\approx$  post at identical token counts. The flat  $\approx 42$  GB allocator reservation is not a memory measurement; continuous-batching efficiency remains future work.

## 8 Discussion and Limitations

The native merger is not lossless for selection: across three models from two architecture families, it rewrites saliency in a text-hostile direction. RBM is therefore a robust default, not a universal winner. It retains dense OCR better than the compared FastV settings (+16.0/+8.5 pp on the two Qwen models), while FastV leads on several scene-text/object-centric cells (Table 2).

**Architecture and mechanism scope.** M1–M3 are established on Qwen3-VL; Qwen2.5-VL retains an ordering confound, and InternVL3 provides accuracy-level evidence. Models without a native merger are out of scope. M1/M2 use a Sobel proxy on  $n = 64$  samples per benchmark and M3 uses  $n = 200$ ; the GQA null has no equivalence margin, and multiple comparisons have no family-wise correction.

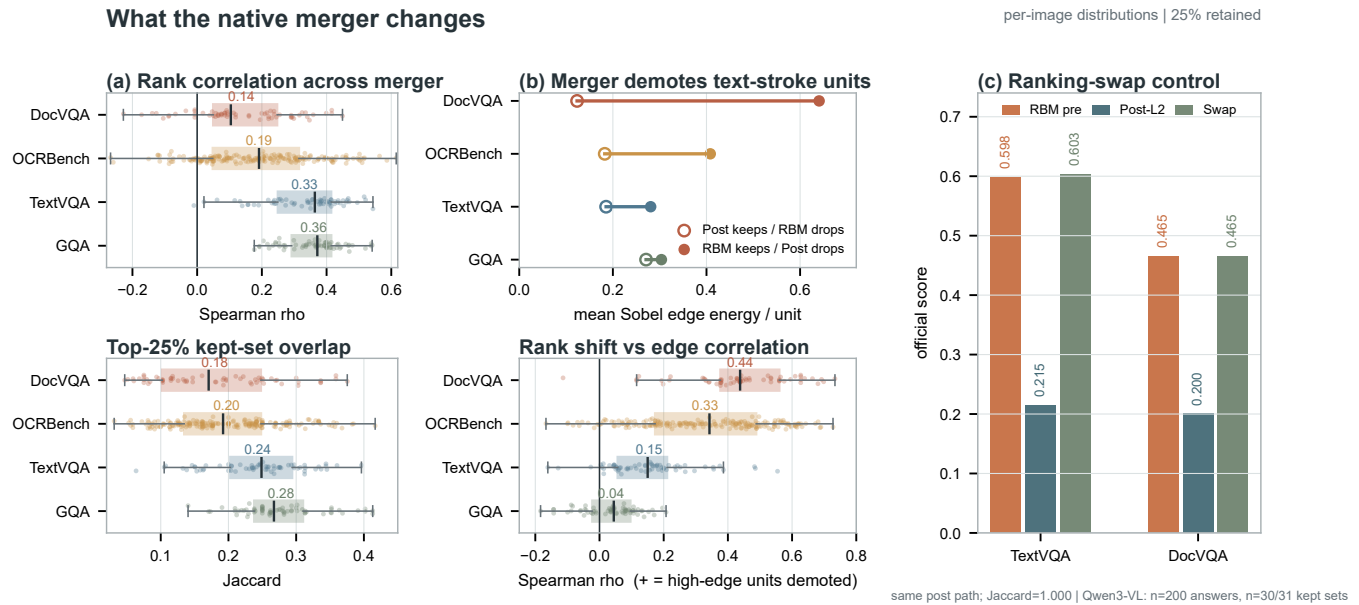
**Evaluation scope.** FastV uses HF eager whereas the other methods use vLLM; anchors support accuracy comparison, but throughput is vLLM-only. Some baseline cells use  $n = 200$ , as marked. Metric-level bootstrap intervals and paired permutation tests are primary; McNemar covers additional binary outcomes.

**Configuration scope.** DocVQA pixel caps differ (4 M for InternVL3, 600k for Qwen baseline cells), so cross-model absolute scores are incomparable. Low-resolution Qwen2.5-VL reverses the shallow-stage direction; the headline is the native-resolution full split. RBM also remains lossy against no compression on InternVL3 DocVQA/OCR (−19.4 pp/−99 points), despite beating post.

## 9 Conclusion

We isolate visual-token selection stage under iso-model, iso-selector, and iso-budget control. Across three models from two architecture families, pre-merger ranking consistently protects text-dense accuracy, with gaps that grow under compression; a ranking-swap control attributes the Qwen3-VL gap to merger-induced rank changes. The resulting training-free method, RBM, is a robust OCR-oriented default rather than a universal winner: query-conditioned FastV





**Figure 3: What the native merger changes. (a) Pre/post L2 rankings are only moderately preserved, measured by Spearman correlation and top-25% kept-set Jaccard across DocVQA ( $n = 64$ ), OCRBench ( $n = 199$ ), TextVQA ( $n = 64$ ), and GQA ( $n = 64$ ). (b) Rank shifts are associated with text-stroke edge energy: units with stronger high-frequency structure are demoted more after merging. (c) In the Qwen3-VL ranking-swap control, post-merger features use the pre-merger order while the kept set remains identical (Jaccard = 1.000), isolating ordering from token identity on  $n = 200$  TextVQA and DocVQA samples.**

remains stronger on several scene-text/object-centric settings. RBM adds no stage-specific serving cost, and six prespecified Qwen3-VL extensions fail to improve over their stronger constituents. Code will be released.

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## Supplementary Material

### Rank Before You Merge: A Stage Law for Training-Free Visual Token Compression in Merger-Equipped Vision-Language Models

Every number is labeled with its source digest (experiments/\*.md) or run-JSON path. Tables S1–S3/S7 support § 5 of the body; S4–S6 support § 4 (S4b cascade counts support § 6); S8 supports § 3; S9 is the per-cell reproducibility index, with the full per-cell listing (cell / n / official / ptid / skip / JSON path) in S9.1–S9.4; S10 is a qualitative same-image pre/post selection comparison rendered from measured Qwen3-VL-8B merger L2 (four panels each, no paired question).

#### S1. M1 — merger ranking reshuffle, full table (Qwen3-VL-8B, seed-0, $n = 64$ /benchmark)

Per image, Spearman correlation of the pre ranking (merger-input unit L2) with the post ranking (merged-token L2) over all units, plus top-25% kept-set overlap. Source: mechanism\_verification\_report § 1 (M1 campaign; released with code) via drafts/paper\_v4.md § 4 (measured under the experiments/j3\_mechanism\_crossarch.md campaign). The Kendall- $\tau$  column is taken from the same M1 table in drafts/paper\_v4.md § 4.1.

| Benchmark         | Spearman $\rho$ | Jaccard@25% | Kendall $\tau$ |
|-------------------|-----------------|-------------|----------------|
| DocVQA (document) | 0.137           | 0.180       | 0.094          |
| TextVQA (scene)   | 0.332           | 0.243       | 0.227          |
| GQA (object)      | 0.360           | 0.278       | 0.249          |

**Reading:** the reshuffle is largest on the most text-dense benchmark and smallest (but still substantial) on object-centric GQA —  $\rho = 0.137 < 0.332 < 0.360$ ; Jaccard =  $0.180 < 0.243 < 0.278$ . Chance-level Jaccard at  $\kappa = 25\%$  is 0.25 (overlap is below random for DocVQA, i.e. actively anti-correlated in the decision band).

#### S2. M2 — demotion directionality vs text-stroke energy, full table (same sample as S1)

rank\_shift = post\_rank – pre\_rank (+  $\Rightarrow$  demoted); per-unit Sobel edge energy as the text-stroke proxy. Source: mechanism\_verification\_report § 2 (M2 campaign; released with code) via drafts/paper\_v4.md § 4. The per-benchmark Sobel means and median fractions (TextVQA, GQA) are taken from the same M2 table in drafts/paper\_v4.md § 4.2.

| Benchmark | $\rho$ | Sobel drop | Sobel reverse | demoted > median |
|-----------|--------|------------|---------------|------------------|
| DocVQA    | +0.439 | 0.641      | 0.124         | 92% (vs 35%)     |
| TextVQA   | +0.155 | 0.281      | 0.186         | 72% (vs 53%)     |
| GQA       | +0.036 | 0.304      | 0.271         | 58% (vs 53%)     |

**Reading:** the merger preferentially demotes high-edge (text-stroke) units  $\sim 12\times$  more strongly on documents than on object scenes; on DocVQA the units post drops that pre keeps are the highest-edge of all (92% above the per-image median vs 35% for the reverse group) — post-merger selection is systematically anti-text on documents. Directionality hypothesis (merger trained on natural content attenuates high-frequency stroke energy): testable hypothesis, not a measured result. Second text-stroke proxy (Laplacian variance): future work (single-proxy disclosure).

#### S3. Retention curve — full number strings ( $n = 200$ , both Qwen models)

pre–post  $\Delta$  at 75% / 50% / 25% retention (the body § 5.4 prints endpoints only). Source: experiments/j8\_ablations.md via drafts/paper\_v4.md § 5.6.

| Model         | Benchmark | $\Delta$ @75% | $\Delta$ @50% | $\Delta$ @25%   |
|---------------|-----------|---------------|---------------|-----------------|
| Qwen3-VL-8B   | TextVQA   | +8.7 pp       | +29.0 pp      | <b>+38.4 pp</b> |
| Qwen3-VL-8B   | DocVQA    | –1.3 pp       | +5.9 pp       | <b>+24.3 pp</b> |
| Qwen2.5-VL-7B | TextVQA   | +7.0 pp       | +15.3 pp      | <b>+26.1 pp</b> |
| Qwen2.5-VL-7B | DocVQA    | –2.1 pp       | –1.4 pp       | <b>+11.0 pp</b> |

**Reading:** the two stages are indistinguishable under shallow compression (DocVQA post even leads 1–2 pp within noise at 75%); the pre-merger lead emerges and widens monotonically with depth. Same shallow regime as the Qwen2.5-VL 600k-cap reversal (body § 5.4 boundary; § 8 item 3). Data source for Fig. 3.

#### S4. Primary paired McNemar statistics — per cell (full splits, both Qwen models)

Primary test = paired McNemar  $z$  on per-sample correctness,  $z = (b - c) / \sqrt{b + c}$ , with  $b/c$  the pre-only / post-only correct counts; positive  $z$  = pre-merger lead, negative = post-merger lead (GQA only). Cross-check = independent-binomial  $z$  (SE  $\sqrt{se_{pre}^2 + se_{post}^2}$ ); “agree %” = per-question agreement of the recorded correctness indicators. Source: drafts/paper\_v4.md § 5.2 paired-test table (per-sample rescore of runs/full\_matrix/j7\_\*.full.json, paired by question id, restricted to ids answered in both arms; 200/200 ground-truth self-test) + experiments/j7\_main\_table.md.

$\dagger$  the only significant post-stage lead. These GQA cells use the deepstack[0]-input (layer-8) pre ranking; a pure-stage control that ranks at the main-merger input (pre-final) gives 0.0 pp on GQA (body § 4.2, S7b), so the apparent post>pre is a layer-8 ranking artifact, not a true stage reversal. GQA official word-normalized exact-match rescore gives a concordant McNemar  $z = 8.1/7.1$  (Qwen3-VL / Qwen2.5-VL). Smallest text-dense  $\Delta_{25} |z| = 14.6$  (Qwen2.5-VL OCR-Bench); smallest text-dense  $\Delta_{12.5} |z| = 15.0$  (Qwen2.5-VL OCR-Bench); smallest  $|z|$  anywhere = 5.7 (Qwen2.5-VL GQA). On OCR-Bench @25% the independent-binomial cross-check (18.2 / 14.7) marginally exceeds McNemar (16.7 / 14.6) because the binary correctness indicator coarsens the /1000 item score (body footnote e); McNemar remains the primary test and is  $\geq$ -consistent on every other cell.

#### S4b. Cascade two-stage gate — paired McNemar counts (Qwen3-VL-8B, $n = 200$ , official rescore)

Cascade gate (body Table 4 / § 6): single budget, Stage 1 = pre-merger RBM keeping fraction  $X$  of merger-input units, Stage 2 = FastV at LLM layer 2 dropping fraction  $Y$  of the survivors; total keep =  $X \cdot (1 - Y)$ . Three arms (pre-alone / fastv-alone / cascade) are compared pairwise on identical images ( $n_{common}$  matched; mean post-merger token count identical across all three arms in every cell, so the comparison is iso-budget). cas\_only / pre\_only / fst\_only = per-sample counts correct in only the named arm of the pair. Verdict = NO-GO (cascade fails the Pareto rule on all 8 budget $\times$ benchmark cells). Source: runs/cascade/gate\_result.json (mcnemar\_cas\_vs\_pre / mcnemar\_cas\_vs\_fst); experiments/cascade\_gate.md.

**Reading:** cascade sits strictly between the two single methods and is dominated by the better arm on every cell — pre-alone on OCR-Bench ( $z = -5.15 / -5.01$  vs pre at 25% / 12.5%); the FastV stage destroys exactly the raw-patch information Stage 1 was built to protect) and FastV-alone on TextVQA / GQA ( $z = -2.34 / -2.43$  vs fst at 25%). The single positive cas-vs-pre  $z$  (DocVQA 25%, +2.07) is paired with a loss to FastV (–1.50), so the “strictly beats both arms” condition holds nowhere. Two training-free selectors composed in series do not complement (body § 6).

#### S4c. Selector-scorer and stage-router gates (Qwen3-VL-8B, official rescore)

**S4c.1 Frequency-aware selector** (body § 6e): the mechanism-side probe of body § 3,  $s_{freq}(u) = \alpha z(\|f_u\|_2) + \beta z(\text{Var}_u)$ , tuned in two grid rounds on TextVQA<sub>500</sub> at 25% retention ( $\alpha=1, \beta=0.6$ ; official VQA-acc). The variance term helps only the most text-dense cell and only under aggressive compression; everywhere else it is neutral or negative, so no no-regression-across-four-benchmarks gate passes. Source: experiments/freq\_aware/results.json + experiments/freq\_aware/grid.json (scripts/freq\_adaptive\_driver.py).

**S4c.2 Adaptive stage router** (body § 6f): query-blind per-image workload detector on merger-input features — high-frequency tail ratio  $r_{HF} = \Pr[\text{Var}_u > \text{Var} + \sigma_{\text{Var}}]$  and 20-bin L2-entropy  $H(\|f_u\|_2)$  — routing each image to PRE (RBM, rank-before-merge) or POST (FastV, merge-then-rank), both iso-budget.  $\kappa$  is the kept fraction; the winner of a  $r_{HF} \times \tau_{ent}$  grid on TextVQA<sub>500</sub> ( $r_{HF}=0.08, \tau_{ent}=2.0$ ) routes every image to PRE, reproducing plain RBM bit-for-bit: text benchmarks show 0.00 pp regression but GQA gain is also +0.00 pp (below the required +1 pp). The  $r_{HF}=0.13$  threshold does route some scene images to POST but collapses TextVQA (51/320 flipped, –7 pp) while flipping only 22/243 GQA images (0.428 vs RBM 0.438). GQA and TextVQA  $r_{HF}$  distributions overlap (means 0.127 vs 0.142), so no single global threshold reconciles the two acceptance criteria. Source: experiments/adaptive\_stage/results.json + experiments/adaptive\_stage/grid.json. Workload-signal analysis: scripts/adapt\_discriminator\_probe.py.

#### S5. Binomial stderrs + skip accounting (Table 1 Qwen blocks)

$\sigma = \sqrt{p(1-p)/n}$ , greedy decoding (no temperature variance; error bars are binomial only). Source: experiments/j7\_main\_table.md (stderr column) + the Qwen-block footnotes of Table 1.

- Qwen3-VL none DocVQA: re-run at max-model-len 49,152, native resolution, 0/5349 skips (ANLS 0.956).



| Model         | Benchmark | $\kappa$ | $\Delta$ (official) | McNemar $z$ (primary) | $b$ (pre-only) | $c$ (post-only) | agree % | indep-binom $z$ |
|---------------|-----------|----------|---------------------|-----------------------|----------------|-----------------|---------|-----------------|
| Qwen3-VL-8B   | TextVQA   | .25      | +38.4 pp            | <b>+43.0</b>          | 2349           | 183             | 49.4    | 42.3            |
| Qwen3-VL-8B   | DocVQA    | .25      | +24.3 pp            | <b>+34.7</b>          | 1601           | 150             | 67.3    | 27.1            |
| Qwen3-VL-8B   | OCR-Bench | .25      | +363 pts            | <b>+16.7</b>          | 422            | 56              | 52.2    | 18.2            |
| Qwen3-VL-8B   | GQA       | .25      | -2.8 pp $\dagger$   | <b>-8.0</b>           | 834            | 1196            | 83.9    | 4.5             |
| Qwen3-VL-8B   | TextVQA   | .125     | +34.0 pp            | <b>+40.8</b>          | 2101           | 161             | 54.8    | 39.9            |
| Qwen3-VL-8B   | DocVQA    | .125     | +24.9 pp            | <b>+33.1</b>          | 1437           | 127             | 70.8    | 32.1            |
| Qwen3-VL-8B   | OCR-Bench | .125     | +297 pts            | <b>+16.0</b>          | 323            | 25              | 65.2    | 17.8            |
| Qwen2.5-VL-7B | TextVQA   | .25      | +26.1 pp            | <b>+30.8</b>          | 1682           | 309             | 60.2    | 27.3            |
| Qwen2.5-VL-7B | DocVQA    | .25      | +11.0 pp            | <b>+15.6</b>          | 1406           | 693             | 60.8    | 11.6            |
| Qwen2.5-VL-7B | OCR-Bench | .25      | +293 pts            | <b>+14.6</b>          | 347            | 54              | 59.9    | 14.7            |
| Qwen2.5-VL-7B | GQA       | .25      | -2.6 pp $\dagger$   | <b>-5.7</b>           | 892            | 1150            | 83.8    | 4.1             |
| Qwen2.5-VL-7B | TextVQA   | .125     | +27.8 pp            | <b>+33.1</b>          | 1836           | 305             | 57.2    | 29.1            |
| Qwen2.5-VL-7B | DocVQA    | .125     | +21.0 pp            | <b>+25.6</b>          | 1325           | 294             | 69.7    | 23.3            |
| Qwen2.5-VL-7B | OCR-Bench | .125     | +268 pts            | <b>+15.0</b>          | 294            | 26              | 68.0    | 15.9            |

| total keep $\kappa$ | bench     | $n$ | pre    | fst    | cas    | $z$ (cas vs pre) | cas_only | pre_only | $z$ (cas vs fst) | cas_only | fst_only |
|---------------------|-----------|-----|--------|--------|--------|------------------|----------|----------|------------------|----------|----------|
| 25%                 | TextVQA   | 200 | 0.5967 | 0.6800 | 0.5950 | -0.152           | 21       | 22       | -2.335           | 18       | 35       |
| 25%                 | DocVQA    | 200 | 0.4239 | 0.5183 | 0.4779 | +2.066           | 38       | 22       | -1.500           | 26       | 38       |
| 25%                 | OCR-Bench | 181 | 0.5801 | 0.1713 | 0.3978 | -5.154           | 4        | 37       | +6.112           | 43       | 2        |
| 25%                 | GQA       | 200 | 0.4150 | 0.4900 | 0.4050 | -0.333           | 17       | 19       | -2.429           | 16       | 33       |
| 12.5%               | TextVQA   | 200 | 0.5000 | 0.4450 | 0.4050 | -2.540           | 21       | 41       | -0.896           | 27       | 34       |
| 12.5%               | DocVQA    | 200 | 0.2980 | 0.2779 | 0.2070 | -2.012           | 31       | 49       | -2.064           | 16       | 30       |
| 12.5%               | OCR-Bench | 181 | 0.3591 | 0.0884 | 0.1713 | -5.013           | 6        | 40       | +3.441           | 17       | 2        |
| 12.5%               | GQA       | 200 | 0.3950 | 0.4550 | 0.3500 | -1.286           | 20       | 29       | -3.130           | 12       | 33       |

| bench@ $\kappa$ | freq( $\alpha=1, \beta=0.6$ ) | RBM (L2) | FastV (post) | $\Delta$ freq-RBM |
|-----------------|-------------------------------|----------|--------------|-------------------|
| TextVQA @25%    | 0.6173                        | 0.6053   | 0.2073       | +1.20             |
| TextVQA @50%    | 0.6780                        | 0.6733   | 0.3873       | +0.47             |
| DocVQA @25%     | 0.4903                        | 0.5063   | 0.2317       | -1.60             |
| DocVQA @50%     | 0.6271                        | 0.6212   | 0.5487       | +0.59             |
| OCR-Bench @25%  | 0.7160                        | 0.7380   | 0.1700       | -2.20             |
| OCR-Bench @50%  | 0.8220                        | 0.8220   | 0.4980       | 0.00              |
| GQA @25%        | 0.4220                        | 0.4380   | 0.4540       | -1.60             |
| GQA @50%        | 0.4860                        | 0.4840   | 0.5000       | +0.20             |

**Table 3: Frequency-aware selector (official metrics,  $n = 500$ ; TextVQA=1.2 pp gain at  $\kappa=25\%$ , baseline-equal elsewhere, strong regression on the other three benchmarks at 25%).**

- OCR-Bench none cells skip **18/1000 (Qwen3-VL)** / **24/1000 (Qwen2.5-VL)** long-OCR images on context overrun, scored 0 (conservative); compressed cells skip  $\leq 5$ . The uncompressed anchor is therefore understated and the relative gaps are conservative.
- Qwen2.5-VL OCR-Bench @25% RBM: 4 M-pixel cap (mean tokens 229.6 vs post 282.0)  $\rightarrow$  its +293-pt gap is conservative (the one iso-token exception in the Qwen block of Table 1, note d).

Per-cell  $\sigma$  values (full splits, computed from the printed official metric  $p$  and full-split  $n$ ; greedy decoding  $\Rightarrow$  binomial only):

$\sigma = \sqrt{p(1-p)/n}$ ; each cell's every text-dense  $\Delta_{25}$  is  $\geq 15\sigma$  of its own cells' pooled stderr, and the full-split GQA post lead (+2.8/+2.6 pp) is  $\approx 5\sigma$  of the pooled GQA stderr ( $\approx 0.0063$ ) — significant but an order of magnitude smaller than any text-dense  $\Delta$ , and reframed by the pre-final pure-stage control (S7b) as a layer-8 ranking artifact rather than a true stage reversal. <sup>a</sup> Qwen3-VL none-DocVQA native-resolution anchor: full-split run partly skipped on context overrun (re-run at max-model-len 49,152 gives ANLS 0.956, 0/5349 skips);  $\sigma$  not printed for the partial native cell. <sup>c</sup> GQA evaluated at  $\kappa = 0.25$  only on the full split (no full-split  $\kappa = 0.125$  cells; body footnote c).

- Qwen3-VL OCR-Bench none = 760/1000 over 982 attempted (18 skips scored 0); compressed @25%: 547 (pre) / 184 (post), skip  $\leq 5$  — ratio 3.0 $\times$ , widening to 6.6 $\times$  at 12.5% (350 vs 53).

## S6. $n = 200$ cross-split consistency (dev-subset vs full-split direction)

The  $n = 200$  dev scope used by Tables 3–4 and §5.3 is direction-consistent with the full splits. Source: experiments/j2\_crossgen\_matrix.md ( $n = 200$  official) + drafts/paper\_v4.md Table A2 (j0a/j2/j4 campaigns).  $\Delta_{25} = \text{pre} - \text{post}$  at  $\kappa = 0.25$ .

**Reading:** all four text-dense cells agree in sign (pre-merger lead) across the two scopes; magnitudes are within run noise (the  $n = 200$  subset SE is  $\approx 8\times$  larger, so the subset is retained only as a direction check, never as the headline). The full Qwen3-VL  $n = 200$  cross-method view (RBM vs FastV vs Pyramid) and the compact Qwen2.5-VL  $n = 200$  cross-generation matrix are printed in paper\_v4 Table A2 and §5.4 Table 2 respectively.

## S7. Selector invariance (second scorer family, Qwen3-VL)

The stage law under a second, non-L2 scorer (global-centroid attention). Source: method\_gate\_report via drafts/paper\_v4.md §5 (j3/j5 campaigns).

**Reading:** invariant in sign and magnitude on Qwen3-VL — the effect is not an L2 artifact. On Qwen2.5-VL the L2 sign is invariant but the attention proxy degrades (proxy-family specificity, not a counter-example); L2 remains the paper's selector, frozen as plain RBM (§3.2).

## S7b. Pure-stage confound control — pre-final (Qwen3-VL-8B, $n = 200$ )

The body §4.2 pre-final control isolates the *stage* from the feature-space confound of the original pre (deepstack[0]-input, layer 8) vs post (main-merger output) contrast. pre-final ranks L2 at the main-merger's *own input* (final ViT-block features) and selects the top- $\kappa$  merge-units from the visual output; the deepstack mergers run untouched. The only difference from post is then before vs after the lossy 2 $\times$ 2 main merger — a pure stage variable. Iso-token verified (prompt token ids match 200/200 per benchmark for pre-final vs post). Source: experiments/p0-3\_prefinal\_control.md; per-sample JSONs in runs/qwen3\_prefinal\_control/.

**Reading:** on text-dense benchmarks pre-final still leads post by a large margin (the text-dense stage effect is a true stage effect, not a deepstack[0]-input vs main-merger-output feature-space artifact). On GQA pre-final equals post (0.0 pp): there is no stage effect on object-QA, and the original pre-vs-post apparent post>pre (-2.8 pp Qwen3-VL, -2.6 pp Qwen2.5-VL; S4) was an artifact of the deepstack[0]-input layer-8 ranking being poorly suited to object-QA, not a true stage reversal. Consistent with the mechanism: the merger does not preferentially damage object features (M2

| Model         | Benchmark | $n$   | none           | pre@25% | post@25% | pre@12.5%      | post@12.5%     |
|---------------|-----------|-------|----------------|---------|----------|----------------|----------------|
| Qwen3-VL-8B   | TextVQA   | 5000  | 0.0051         | 0.0069  | 0.0059   | 0.0071         | 0.0048         |
| Qwen3-VL-8B   | DocVQA    | 5349  | — <sup>a</sup> | 0.0068  | 0.0058   | 0.0065         | 0.0042         |
| Qwen3-VL-8B   | OCR-Bench | 1000  | 0.0135         | 0.0157  | 0.0123   | 0.0151         | 0.0071         |
| Qwen3-VL-8B   | GQA       | 12578 | 0.0043         | 0.0044  | 0.0045   | — <sup>c</sup> | — <sup>c</sup> |
| Qwen2.5-VL-7B | TextVQA   | 5000  | 0.0049         | 0.0065  | 0.0070   | 0.0069         | 0.0066         |
| Qwen2.5-VL-7B | DocVQA    | 5349  | 0.0030         | 0.0066  | 0.0068   | 0.0068         | 0.0059         |
| Qwen2.5-VL-7B | OCR-Bench | 1000  | 0.0122         | 0.0158  | 0.0122   | 0.0149         | 0.0079         |
| Qwen2.5-VL-7B | GQA       | 12578 | 0.0044         | 0.0044  | 0.0044   | — <sup>c</sup> | — <sup>c</sup> |

| Model      | Bench   | full $\Delta_{25}$ | $n = 200$ $\Delta_{25}$ | sign  |
|------------|---------|--------------------|-------------------------|-------|
| Qwen3-VL   | TextVQA | +38.4 pp           | +38.3 pp                | ✓ pre |
| Qwen3-VL   | DocVQA  | +24.3 pp           | +26.5 pp                | ✓ pre |
| Qwen2.5-VL | TextVQA | +26.1 pp           | +32.0 pp                | ✓ pre |
| Qwen2.5-VL | DocVQA  | +11.0 pp           | +18.8 pp                | ✓ pre |

| Selector | Benchmark           | pre          | post        | $\Delta$                |
|----------|---------------------|--------------|-------------|-------------------------|
| L2       | TextVQA             | 0.598–0.605  | 0.215–0.222 | $\approx +38$ pp        |
| GC-attn  | TextVQA             | <b>0.553</b> | 0.200       | <b>+35.3 pp</b>         |
| GC-attn  | TextVQA/DocVQA (2n) | —            | —           | <b>+35.5 / +24.4 pp</b> |

| Benchmark (metric) | none  | pre-final    | post  | $\Delta$ (paired mean $\pm$ SE) | verdict         |
|--------------------|-------|--------------|-------|---------------------------------|-----------------|
| TextVQA (VQA-acc)  | 0.858 | <b>0.515</b> | 0.215 | +30.0 pp (+0.300 $\pm$ 0.039)   | stage effect    |
| DocVQA (ANLS)      | 0.976 | <b>0.326</b> | 0.200 | +12.5 pp (+0.125 $\pm$ 0.032)   | stage effect    |
| GQA (exact match)  | 0.605 | 0.450        | 0.450 | 0.0 pp (+0.000 $\pm$ 0.032)     | no stage effect |

$\rho = 0.04$  on GQA, S2). pre-final tested on Qwen3-VL only (the deepstack confound is Qwen3-specific; Qwen2.5-VL shares the architecture but was not re-tested).

## S8. Merger tap points and M-RoPE configuration details (supports §3)

- **Qwen3-VL-8B**: the first deepstack merger has index 0 and the visual indexes are [8, 16, 24], so unit scores use **ViT block-8 outputs** (intermediate depth); the single resulting mask is shared by all mergers, including the main merger that consumes the final-block output. Runner diagnostic mask\_computed\_at = 'deepstack\_0' on every run. Source: experiments/r1\_1\_swap\_jaccard.md Part B (three-evidence verification: call order in Qwen3VLVisionModel.forward, model config, run diagnostics).
- **Qwen2.5-VL-7B**: no deepstack; scores are computed on the **main merger input (final ViT block output)**. mask\_computed\_at = 'main'.
- **InternVL3-8B**: the pixel-shuffle downsampling stage is the merger; L2 taps its input.
- **M-RoPE fix**: Qwen2.5-VL uses a block M-RoPE layout and Qwen3-VL an interleaved one; under pruning the position cursor must advance by the actual surviving count. A family-scoped fix (bit-degrading to the original at full retention) makes the Qwen2.5-VL compression path well-formed; the block-vs-interleaved contrast is itself a diagnostic of why naive pruning collapses one generation and is tolerated by the other.
- **Table 2 / r2c harness**: HF transformers eager attention, --mropenative (family-agnostic correct positions) on all same-scope cells; the Qwen2.5-VL  $\times$  pre vllm-mimic positional degeneration (repeated "addCriterion..." outputs  $\rightarrow$  pseudo-zero scores) was detected and replaced before any reported number; on Qwen3-VL native vs mimic give 0.575 vs 0.525 (same ordering). Source: experiments/r2c\_rbm\_scope.md.

## S9. Per-cell reproducibility index (run JSONs; gitignored, released with code)

### S9.1 InternVL3-8B full-split cells (Table 1; 16/16)

Model OpenGVLab/InternVL3-8B, vLLM 0.19, bf16, eager attention, greedy;  $r = \text{TO-TAL drop}$  (0.750  $\rightarrow \kappa = 25\%$ , 0.875  $\rightarrow \kappa = 12.5\%$ ); skip = context-overflow cells scored 0. Source: runs/internvl3/internvl3\_official\_summary.json (+ per-cell \*\_full.json, each carries n\_skipped; all 0 here). Digest: experiments/internvl3\_main\_matrix.md.  
(All paths relative to runs/internvl3/. Pre-merger leads on every text-dense benchmark at both budgets — e.g. DocVQA ANLS 0.7284 vs 0.382 at  $\kappa = 25\%$ ; GQA is

the near-null object-centric case, 0.5993 vs 0.6031 — replicating the stage law on the pixel-shuffle merger design.)

### S9.2 Table 2 FastV-k3 same-scope cells (8/8)

FastV one-shot attention pruning at LLM layer  $K = 3$ , iso-budget  $r = 0.75$  ( $\kappa = 25\%$ ); HF-transformers 4.57.6 eager harness, official metrics. Official values are the rescored results in experiments/r2b\_fastv\_k3.md; n, ptid, skip, engine, and JSON paths are audited against the per-cell runs/r2\_same\_scope/r2b\_\*.json files. The harness's raw acc field is not used as the official score. (Filenames marked \_n500 for the Qwen3-VL DocVQA / OCR-Bench cells ran at the  $n = 200$  same-scope budget recorded in the JSON n field; OCR-Bench skips are long-OCR context overruns, scored 0.)

### S9.3 Table 2 RBM same-scope cells (5/5)

RBM (pre-merger L2) at iso-budget  $r = 0.75$ , HF-transformers eager, --mropenative (family-agnostic correct positions), matched to the FastV-k3 cells above. Official values are the rescored results in experiments/r2c\_rbm\_scope.md; ptid, skip, and paths are audited against the per-cell runs/r2\_same\_scope/r2c\_\*.json files.

### S9.4 Cascade gate cells (Table 4; 24/24, Qwen3-VL-8B)

Three arms  $\times$  two budgets  $\times$  four benchmarks;  $n = n_{\text{common}}$  (matched by question id); official = acc\_official from the paired rescore; ptid = mean post-merger token count (identical across the three arms in every cell). Source: runs/cascade/gate\_result.json (authoritative paired summary) with per-arm run cells runs/cascade/gate\_{pre, fst, cas}\{25, 12\}\_<bench>.json. Digest: experiments/cascade\_gate.md. (McNemar counts for these cells: S4b.)  
(All paths relative to runs/cascade/. OCR-Bench  $n_{\text{common}} = 181$  after 19 long-OCR skips; all other cells  $n = 200$ . missing\_cells: [] — 24/24 present.)

## S10. Qualitative same-image comparison — measured pre- vs post-merger L2 selection

For four representative inputs we render, on the *same* processor-resized image, unit grid, 25% budget and final visual-token count, the original image with its full merge-unit grid, the RBM (pre-merger L2 top-k) kept mask, the post-merger L2 top-k kept mask (same L2 selector, same budget), and a four-state difference map (kept-by-both / RBM-only / post-only / dropped-by-both, encoded by colour and hatch/outline so the states remain distinguishable in greyscale). Each panel reports the measured top-k Jaccard index  $J = |\text{pre} \cap \text{post}| / |\text{pre} \cup \text{post}|$  and the full-ranking Spearman  $\rho$  between the pre- and post-merger L2 orderings (computed from the stored NPZ rank arrays; values cross-checked against the validation report,  $|\Delta| < 10^{-5}$ ). All scores are the target model's own merger-input / merger-output feature L2 norms (Qwen3-VL-8B-Instruct, enforce\_eager, max\_pixels = 1.5M); no CLIP, SigLIP, attention map, or fabricated saliency is used. The FastV layer-2 panel is intentionally absent (no user-supplied question exists for these images; a query-conditioned mask cannot be measured without one) and is recorded as fastv\_status: skipped\_missing\_question.

The consistently low Jaccard (0.25–0.35) and moderate Spearman (0.30–0.48) across all ten captured inputs (four shown; the remaining six ship in the code release) are the image-level signature of the body paper's mechanism result: the learned  $2 \times 2$  merger reshuffles the unit ranking, so selecting before versus after it on the same image with the same selector and budget keeps largely *different* units. The four-state maps show that the disagreement is spatially structured (contiguous RBM-only and post-only regions), not uniform noise, and that the "kept-by-both" intersection is a sparse minority of the budget.

**Packaging note:** this file converts to the ACM MM supplementary PDF/archive; run JSONs ship in the code release (anonymous archive at submission, repository at de-anonymization). Nothing in this supplement identifies authors, institutions, or repositories (double-blind).

| Body location                 | Cells                 | Runs path                                                            | Summary digest                                              |
|-------------------------------|-----------------------|----------------------------------------------------------------------|-------------------------------------------------------------|
| Table 1 (Qwen blocks)         | 26/26                 | runs/full_matrix/j7_*.json                                           | experiments/j7_main_table.md                                |
| Table 1 (InternVL3 block)     | 16/16                 | runs/internvl3/internvl3_*.json +<br>internvl3_official_summary.json | experiments/internvl3_main_<br>matrix.md                    |
| Table 2 FastV-k3              | 8                     | runs/r2_same_scope/r2b_*.json                                        | experiments/r2b_fastv_k3.md                                 |
| Table 2 RBM same-scope        | 5                     | runs/r2_same_scope/r2c_*.json                                        | experiments/r2c_rbm_scope.md                                |
| Table 4 cascade gate          | 24/24                 | runs/cascade/gate_*.json + gate_<br>result.json                      | experiments/cascade_gate.md                                 |
| Table 5 efficiency            | 7 configs             | runs/full_matrix/efficiency/*.<br>eff.json                           | experiments/j6_efficiency.md                                |
| § 5 kept-set Jaccard / swap   | 8                     | runs/r1_1_swap_jaccard/*.json +<br>jaccard_summary.json              | experiments/r1_1_swap_jaccard.md                            |
| § 5.1–5.2 M1/M2               | $n = 64 \times 3$     | mechanism campaign (released with<br>code)                           | mechanism_verification_report via<br>drafts/paper_v4.md § 4 |
| § 5.4 retention curve         | $n = 200 \times 4$    | j8 ablation runs                                                     | experiments/j8_ablations.md                                 |
| § 6 (a)(b)(c)                 | —                     | gate/j5 runs                                                         | experiments/j5_qa_gate_result.md                            |
| router/hybrid/QA-gate         |                       |                                                                      | + paper_v4 § 6                                              |
| § 8 600k reversal             | 3                     | j8b runs                                                             | experiments/j8b_q2v1_docvqa_600k.<br>md                     |
| Engine fairness               | $r=0$ 8/8; none 16/16 | j4 runs                                                              | experiments/j4_step2_fix.md                                 |
| PyramidDrop iso-25% col-lapse | $n = 500$             | j7hf runs                                                            | experiments/j7hf_baselines_n500.<br>md                      |

| cell                 | mode | $r$   | bench    | $n$   | official | metric         | ptid   | skip | JSON                                     |
|----------------------|------|-------|----------|-------|----------|----------------|--------|------|------------------------------------------|
| none_docvqa          | none | 0.000 | docvqa   | 5349  | 0.9221   | ANLS           | 3230.8 | 0    | internvl3_none_docvqa_r0.000_full.json   |
| none_gqa             | none | 0.000 | gqa      | 12578 | 0.6293   | acc-official   | 2557.2 | 0    | internvl3_none_gqa_r0.000_full.json      |
| none_ocrbench        | none | 0.000 | ocrbench | 1000  | 0.852    | Final/1000=852 | 2040.1 | 0    | internvl3_none_ocrbench_r0.000_full.json |
| none_textvqa         | none | 0.000 | textvqa  | 5000  | 0.8338   | VQA-acc        | 2562.9 | 0    | internvl3_none_textvqa_r0.000_full.json  |
| pre_docvqa_r0.750    | pre  | 0.750 | docvqa   | 5349  | 0.7284   | ANLS           | 859.5  | 0    | internvl3_pre_docvqa_r0.750_full.json    |
| pre_docvqa_r0.875    | pre  | 0.875 | docvqa   | 5349  | 0.5054   | ANLS           | 464.3  | 0    | internvl3_pre_docvqa_r0.875_full.json    |
| pre_gqa_r0.750       | pre  | 0.750 | gqa      | 12578 | 0.5993   | acc-official   | 690.1  | 0    | internvl3_pre_gqa_r0.750_full.json       |
| pre_ocrbench_r0.750  | pre  | 0.750 | ocrbench | 1000  | 0.753    | Final/1000=753 | 555.4  | 0    | internvl3_pre_ocrbench_r0.750_full.json  |
| pre_textvqa_r0.750   | pre  | 0.750 | textvqa  | 5000  | 0.789    | VQA-acc        | 690.5  | 0    | internvl3_pre_textvqa_r0.750_full.json   |
| pre_textvqa_r0.875   | pre  | 0.875 | textvqa  | 5000  | 0.723    | VQA-acc        | 378.4  | 0    | internvl3_pre_textvqa_r0.875_full.json   |
| post_docvqa_r0.750   | post | 0.750 | docvqa   | 5349  | 0.382    | ANLS           | 859.5  | 0    | internvl3_post_docvqa_r0.750_full.json   |
| post_docvqa_r0.875   | post | 0.875 | docvqa   | 5349  | 0.2451   | ANLS           | 464.3  | 0    | internvl3_post_docvqa_r0.875_full.json   |
| post_gqa_r0.750      | post | 0.750 | gqa      | 12578 | 0.6031   | acc-official   | 690.1  | 0    | internvl3_post_gqa_r0.750_full.json      |
| post_ocrbench_r0.750 | post | 0.750 | ocrbench | 1000  | 0.321    | Final/1000=321 | 555.4  | 0    | internvl3_post_ocrbench_r0.750_full.json |
| post_textvqa_r0.750  | post | 0.750 | textvqa  | 5000  | 0.4148   | VQA-acc        | 690.5  | 0    | internvl3_post_textvqa_r0.750_full.json  |
| post_textvqa_r0.875  | post | 0.875 | textvqa  | 5000  | 0.3064   | VQA-acc        | 378.4  | 0    | internvl3_post_textvqa_r0.875_full.json  |

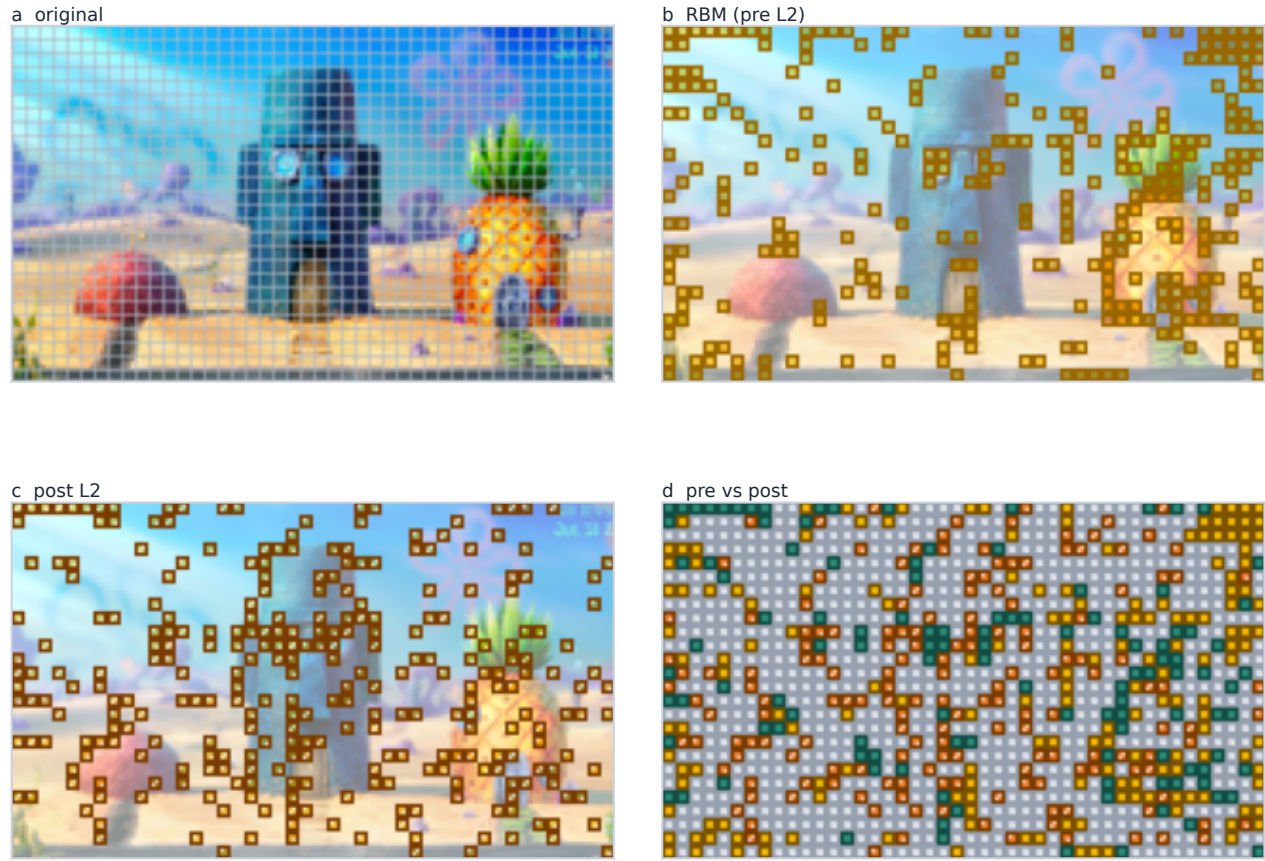
| cell                                        | model      | bench    | $n$   | official | ptid  | skip | engine   | JSON                                             |
|---------------------------------------------|------------|----------|-------|----------|-------|------|----------|--------------------------------------------------|
| r2b_qwen3vl_fastv_k3_textvqa_r0.75_full5000 | Qwen3-VL   | textvqa  | 5000  | 0.7771   | 213.7 | 5    | hf eager | r2b_qwen3vl_fastv_k3_textvqa_r0.75_full5000.json |
| r2b_qwen3vl_fastv_k3_gqa_r0.75_full12578    | Qwen3-VL   | gqa      | 12578 | 0.5376   | 96.8  | 0    | hf eager | r2b_qwen3vl_fastv_k3_gqa_r0.75_full12578.json    |
| r2b_qwen3vl_fastv_k3_docvqa_r0.75_n500      | Qwen3-VL   | docvqa   | 200   | 0.5863   | 176.0 | 0    | hf eager | r2b_qwen3vl_fastv_k3_docvqa_r0.75_n500.json      |
| r2b_qwen3vl_fastv_k3_ocrbench_r0.75_n500    | Qwen3-VL   | ocrbench | 200   | 0.415    | 114.8 | 19   | hf eager | r2b_qwen3vl_fastv_k3_ocrbench_r0.75_n500.json    |
| r2b_qwen2vl_fastv_k3_textvqa_r0.75_n200     | Qwen2.5-VL | textvqa  | 200   | 0.7467   | 282.1 | 0    | hf eager | r2b_qwen2vl_fastv_k3_textvqa_r0.75_n200.json     |
| r2b_qwen2vl_fastv_k3_gqa_r0.75_n200         | Qwen2.5-VL | gqa      | 200   | 0.475    | 126.8 | 0    | hf eager | r2b_qwen2vl_fastv_k3_gqa_r0.75_n200.json         |
| r2b_qwen2vl_fastv_k3_docvqa_r0.75_n200      | Qwen2.5-VL | docvqa   | 200   | 0.4852   | 231.4 | 0    | hf eager | r2b_qwen2vl_fastv_k3_docvqa_r0.75_n200.json      |
| r2b_qwen2vl_fastv_k3_ocrbench_r0.75_n200    | Qwen2.5-VL | ocrbench | 200   | 0.285    | 143.6 | 20   | hf eager | r2b_qwen2vl_fastv_k3_ocrbench_r0.75_n200.json    |

| cell                                | model      | bench    | $n$ | official | ptid  | skip | engine   | JSON                                     |
|-------------------------------------|------------|----------|-----|----------|-------|------|----------|------------------------------------------|
| r2c_qwen2vl_pre_r0.75_textvqa_n200  | Qwen2.5-VL | textvqa  | 200 | 0.6683   | 282.1 | 0    | hf eager | r2c_qwen2vl_pre_r0.75_textvqa_n200.json  |
| r2c_qwen2vl_pre_r0.75_gqa_n200      | Qwen2.5-VL | gqa      | 200 | 0.520    | 126.8 | 0    | hf eager | r2c_qwen2vl_pre_r0.75_gqa_n200.json      |
| r2c_qwen2vl_pre_r0.75_docvqa_n200   | Qwen2.5-VL | docvqa   | 200 | 0.5062   | 231.4 | 0    | hf eager | r2c_qwen2vl_pre_r0.75_docvqa_n200.json   |
| r2c_qwen2vl_pre_r0.75_ocrbench_n200 | Qwen2.5-VL | ocrbench | 200 | 0.370    | 143.6 | 20   | hf eager | r2c_qwen2vl_pre_r0.75_ocrbench_n200.json |
| r2c_qwen3vl_pre_r0.75_ocrbench_n200 | Qwen3-VL   | ocrbench | 200 | 0.575    | 114.8 | 19   | hf eager | r2c_qwen3vl_pre_r0.75_ocrbench_n200.json |



| arm             | $\kappa$ | bench    | $n$ | official | ptid  | JSON                     |
|-----------------|----------|----------|-----|----------|-------|--------------------------|
| pre-alone (RBM) | 25%      | textvqa  | 200 | 0.5967   | 212.8 | gate_pre25_textvqa.json  |
| fastv-alone     | 25%      | textvqa  | 200 | 0.6800   | 212.8 | gate_fst25_textvqa.json  |
| cascade         | 25%      | textvqa  | 200 | 0.5950   | 212.8 | gate_cas25_textvqa.json  |
| pre-alone (RBM) | 25%      | docvqa   | 200 | 0.4239   | 176.0 | gate_pre25_docvqa.json   |
| fastv-alone     | 25%      | docvqa   | 200 | 0.5183   | 176.0 | gate_fst25_docvqa.json   |
| cascade         | 25%      | docvqa   | 200 | 0.4779   | 176.0 | gate_cas25_docvqa.json   |
| pre-alone (RBM) | 25%      | ocrbench | 181 | 0.5801   | 114.8 | gate_pre25_ocrbench.json |
| fastv-alone     | 25%      | ocrbench | 181 | 0.1713   | 114.8 | gate_fst25_ocrbench.json |
| cascade         | 25%      | ocrbench | 181 | 0.3978   | 114.8 | gate_cas25_ocrbench.json |
| pre-alone (RBM) | 25%      | gqa      | 200 | 0.4150   | 95.4  | gate_pre25_gqa.json      |
| fastv-alone     | 25%      | gqa      | 200 | 0.4900   | 95.4  | gate_fst25_gqa.json      |
| cascade         | 25%      | gqa      | 200 | 0.4050   | 95.4  | gate_cas25_gqa.json      |
| pre-alone (RBM) | 12.5%    | textvqa  | 200 | 0.5000   | 120.7 | gate_pre12_textvqa.json  |
| fastv-alone     | 12.5%    | textvqa  | 200 | 0.4450   | 120.7 | gate_fst12_textvqa.json  |
| cascade         | 12.5%    | textvqa  | 200 | 0.4050   | 120.7 | gate_cas12_textvqa.json  |
| pre-alone (RBM) | 12.5%    | docvqa   | 200 | 0.2980   | 103.7 | gate_pre12_docvqa.json   |
| fastv-alone     | 12.5%    | docvqa   | 200 | 0.2779   | 103.7 | gate_fst12_docvqa.json   |
| cascade         | 12.5%    | docvqa   | 200 | 0.2070   | 103.7 | gate_cas12_docvqa.json   |
| pre-alone (RBM) | 12.5%    | ocrbench | 181 | 0.3591   | 68.7  | gate_pre12_ocrbench.json |
| fastv-alone     | 12.5%    | ocrbench | 181 | 0.0884   | 68.7  | gate_fst12_ocrbench.json |
| cascade         | 12.5%    | ocrbench | 181 | 0.1713   | 68.7  | gate_cas12_ocrbench.json |
| pre-alone (RBM) | 12.5%    | gqa      | 200 | 0.3950   | 62.3  | gate_pre12_gqa.json      |
| fastv-alone     | 12.5%    | gqa      | 200 | 0.4550   | 62.3  | gate_fst12_gqa.json      |
| cascade         | 12.5%    | gqa      | 200 | 0.3500   | 62.3  | gate_cas12_gqa.json      |

# Same-image iso-budget comparison - df7282e1 (Qwen3-VL-8B measured merger L2)



N=1144 k=286 (25% kept) top-k Jaccard=0.2599 Spearman  $\rho(\text{pre}, \text{post ranks})=0.3141$  unit grid  $26 \times 44 = \text{grid\_thw}/2$  Qwen3-VL-8B measured merger L2

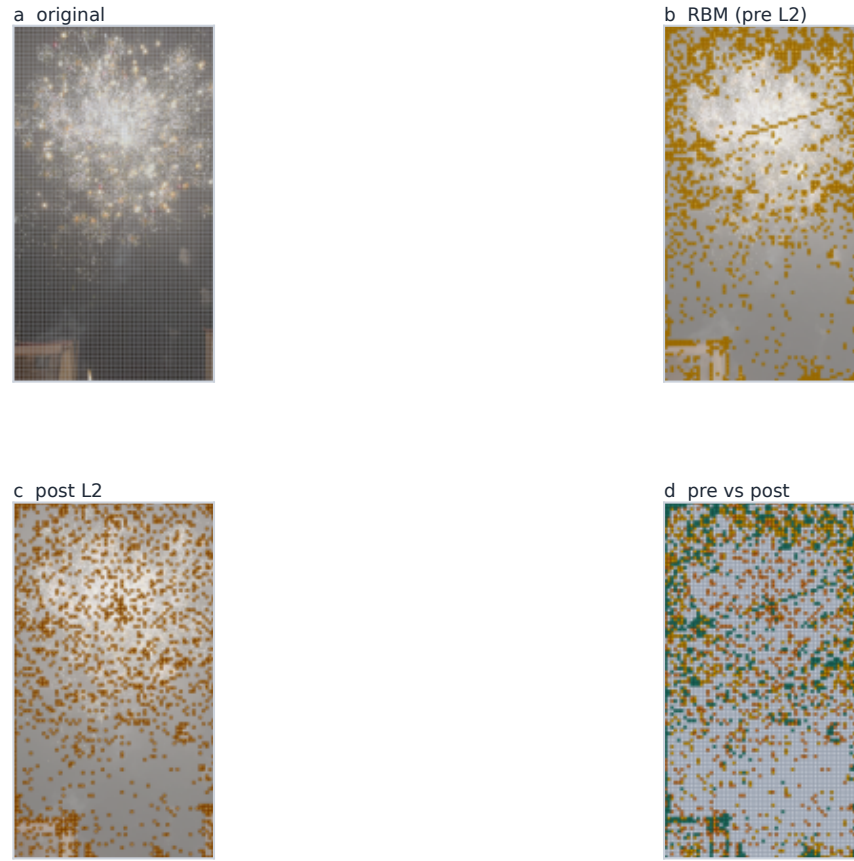
FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).  
 (d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

**Figure 4: Sample df7282e1 ( $N = 1144$ ,  $k = 286$ ,  $J = 0.260$ ,  $\rho = 0.314$ ). Sparse scene; the four-state map shows RBM-only (amber) concentrated on object boundaries while post-only (hatched) scatters over flat regions the merger smoothed.**

## Same-image iso-budget comparison - 69 (Qwen3-VL-8B measured merger L2)



$N=6820$   $k=1705$  (25% kept)  $\text{top-k Jaccard}=0.3478$   $\text{Spearman } \rho(\text{pre, post ranks})=0.4774$   $\text{unit grid } 110 \times 62 = \text{grid\_thw}/2$  Qwen3-VL-8B measured merger L2

FastV: SKIPPED (no user question; inputs/manifest.json absent)

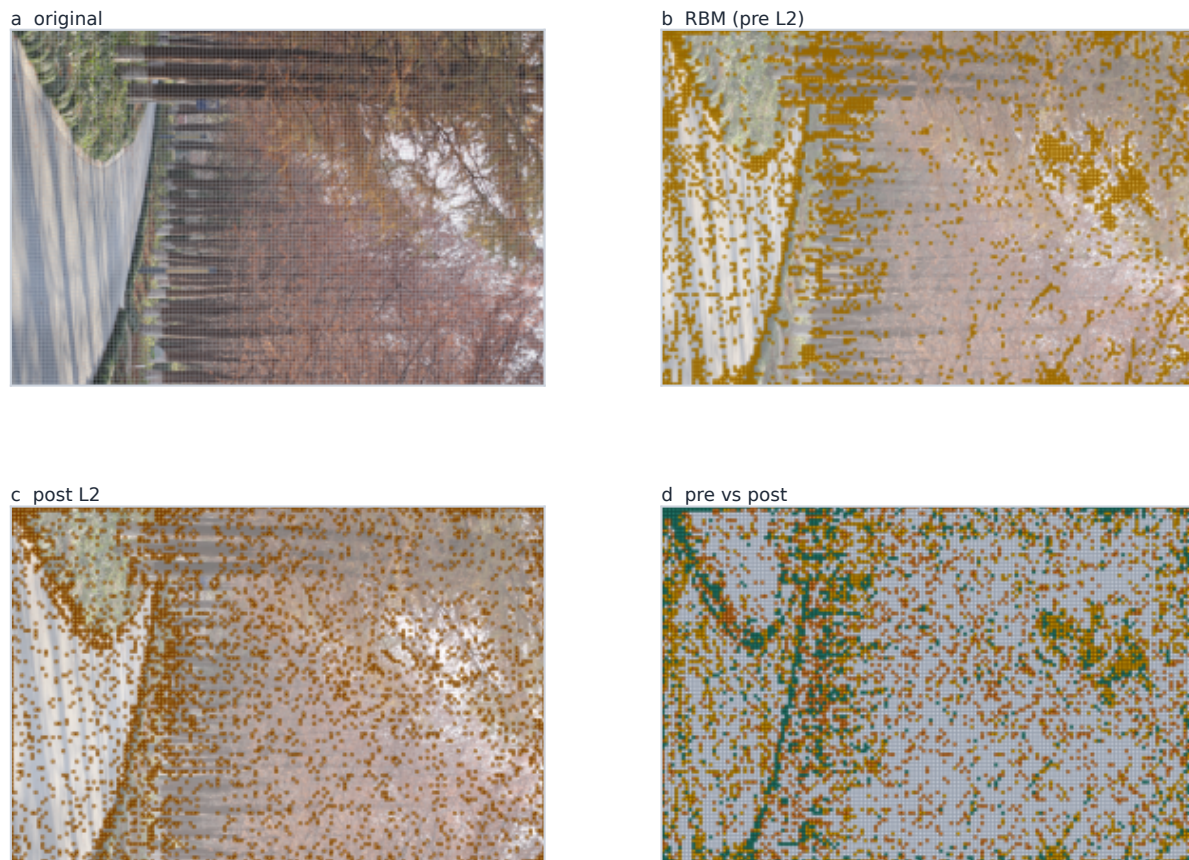
(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).  
(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

**Figure 5: Sample img69 ( $N=6820$ ,  $k=1705$ ,  $J=0.348$ ,  $\rho=0.477$ ). Highest overlap of the four; the merger preserves more of the pre-ranking here, yet two-thirds of the kept set still differs.**



# Same-image iso-budget comparison - 72 (Qwen3-VL-8B measured merger L2)



$N=16224$   $k=4056$  (25% kept)  $\text{top-k Jaccard}=0.2482$   $\text{Spearman } \rho(\text{pre}, \text{post ranks})=0.2999$   $\text{unit grid } 104 \times 156 = \text{grid\_thw}/2$  Qwen3-VL-8B measured merger L2

FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).  
(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

**Figure 6: Sample img72 ( $N=16224$ ,  $k=4056$ ,  $J=0.248$ ,  $\rho=0.300$ ). Textured outdoor scene with a wall edge; the lowest overlap – the merger’s lossy aggregation most strongly reorders the high-frequency edge vs. texture units.**

## Same-image iso-budget comparison - 74 (Qwen3-VL-8B measured merger L2)

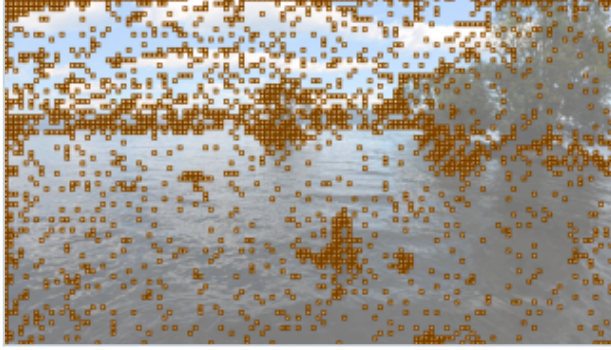
a original



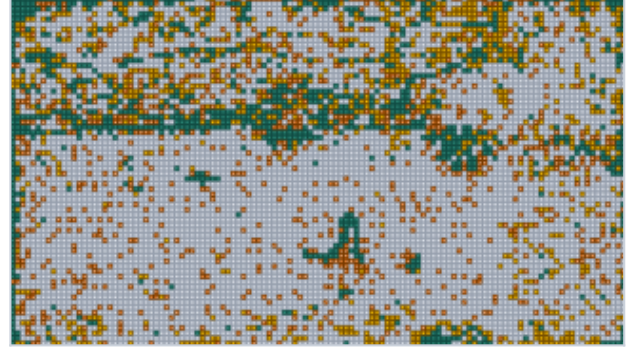
b RBM (pre L2)



c post L2



d pre vs post



$N=8160$   $k=2040$  (25% kept) top-k Jaccard=0.3377 Spearman  $\rho(\text{pre}, \text{post ranks})=0.3963$  unit grid  $68 \times 120 = \text{grid\_thw}/2$  Qwen3-VL-8B measured merger L2

FastV: SKIPPED (no user question; inputs/manifest.json absent)

(a) full unit grid outline (light). (b,c) SAME amber L2 selector & 25% budget; (c) hatched = post (distinguishable without color).

(d) four states kept-by-both / RBM-only / post-only / dropped-by-both, encoded by color + hatch/outline (legend).

■ kept by both (pre & post) ■ RBM-only (pre & ~post) ■ post-only (~pre & post) ■ dropped by both

**Figure 7: Sample img74 ( $N=8160$ ,  $k=2040$ ,  $J=0.338$ ,  $\rho=0.396$ ). Waterscape with horizon; RBM-only units track the water surface detail that the post-merger L2 demotes in favour of merged sky blocks.**